

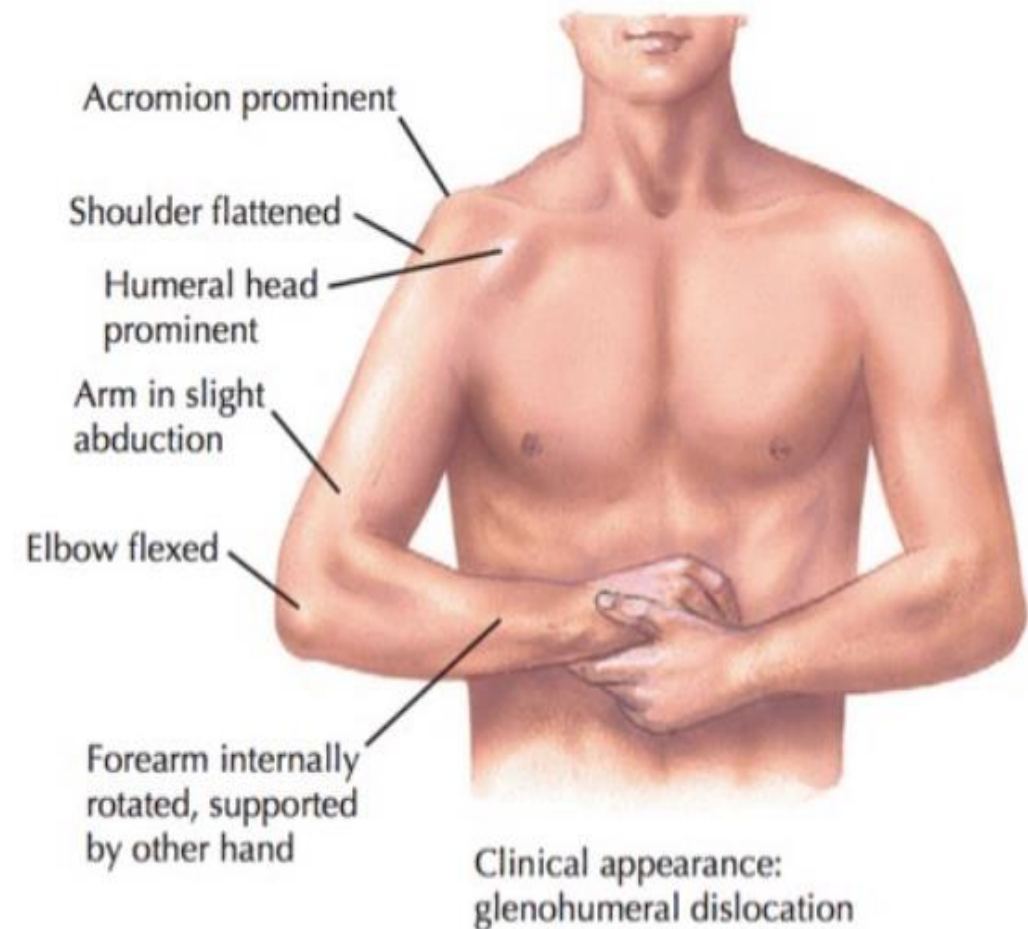
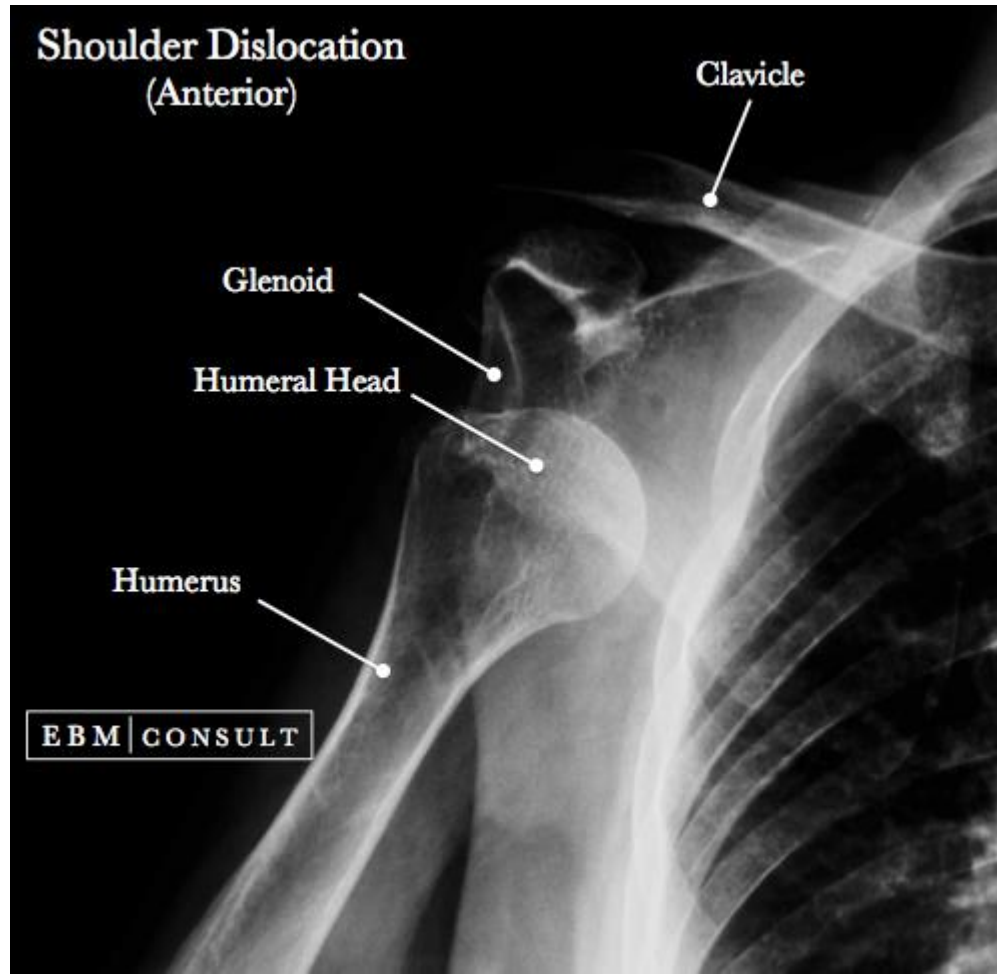
APPLIED ANATOMY OF UL-I-

Dr.Ahmed AL-Baghdadi

Dislocation of shoulder dislocation

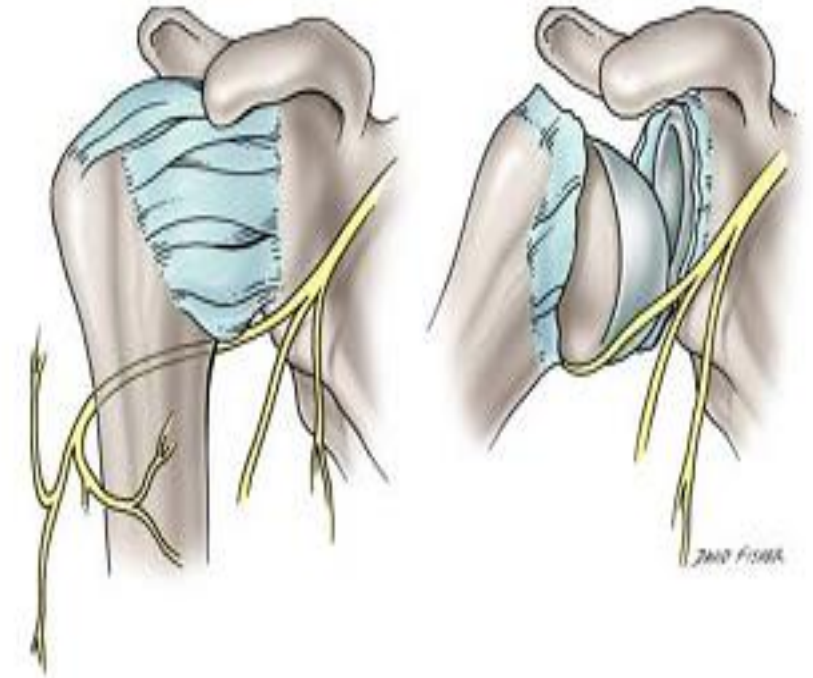
Anterior dislocation of shoulder dislocation

Case scenario: A 22-year-old male presents after a football fall to ground with his arm in **abduction + external rotation**. He now has a **"squared-off" shoulder**, **pain with movement**, and **palpable humeral head anteriorly**



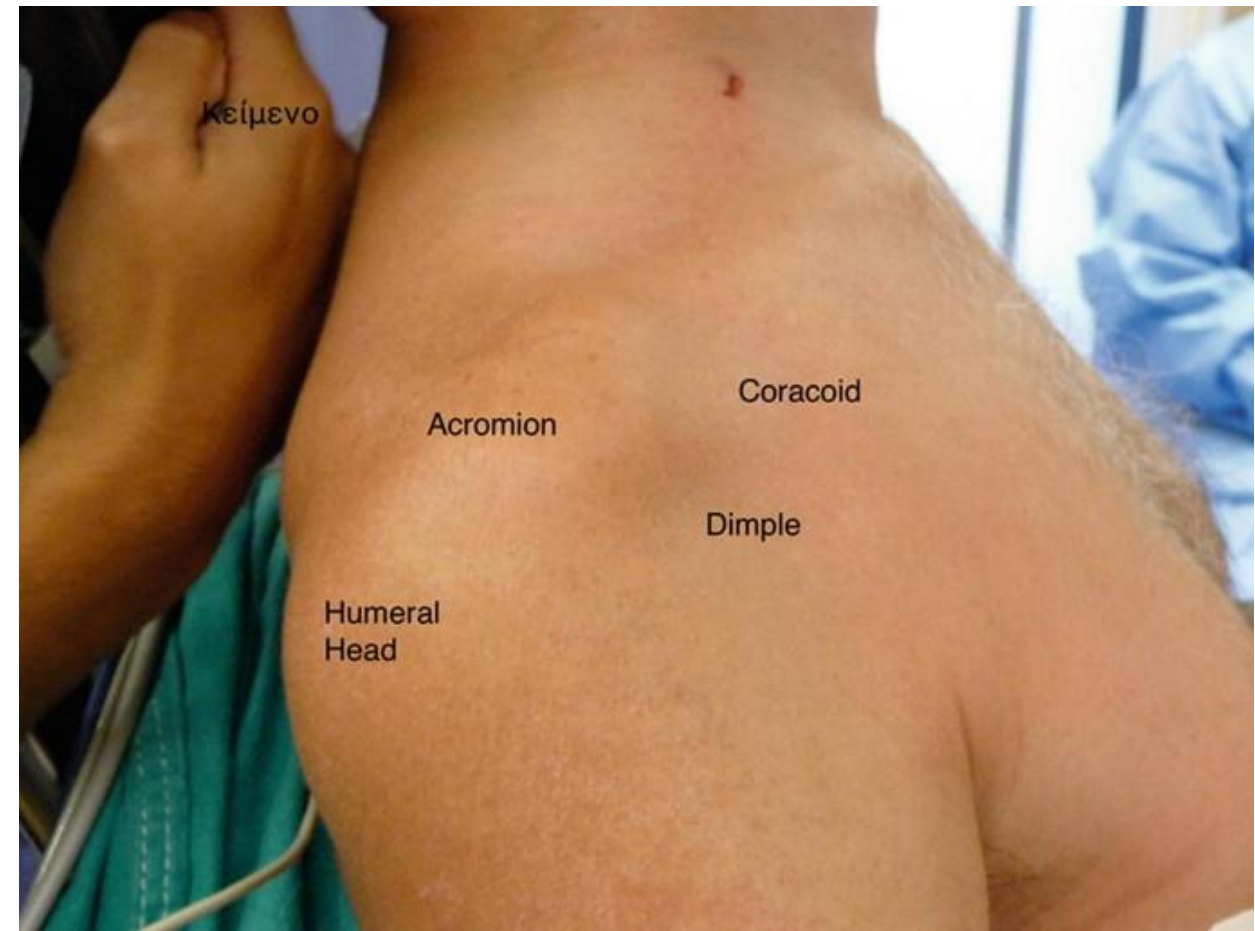
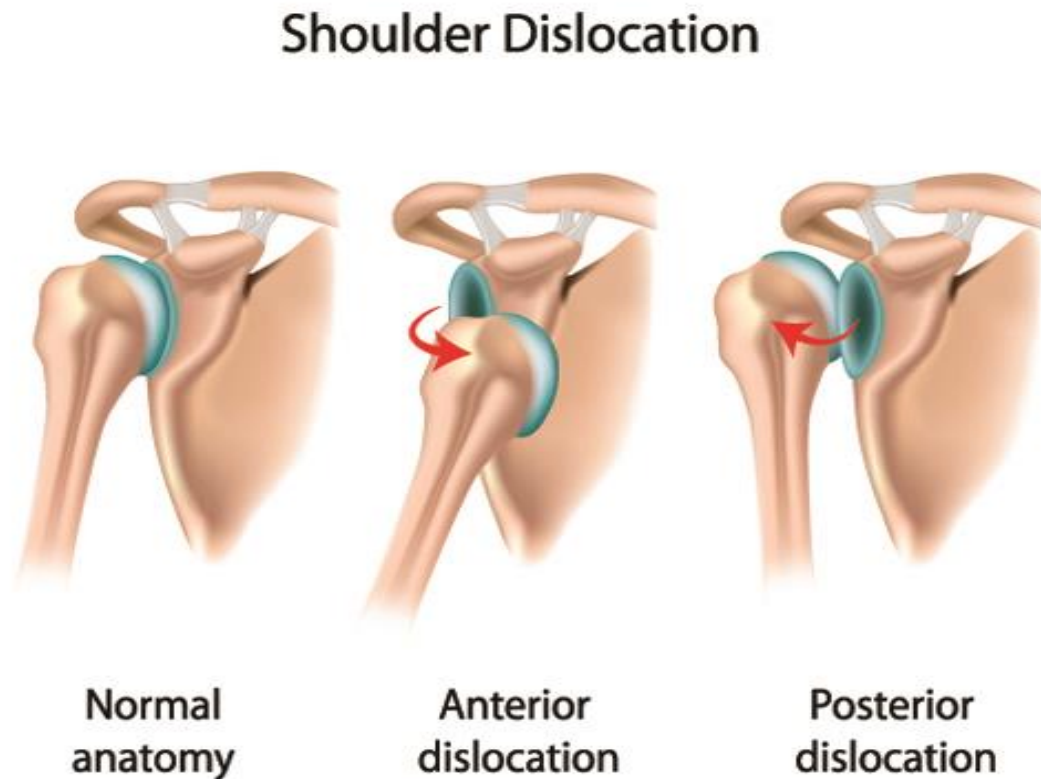
Anterior dislocation of shoulder dislocation

- ❑ Mechanism of trauma?
- ❑ shoulder deformity?
- ❑ Primary restraint failure: anteroinferior labral tear
- ❑ Secondary damage: humeral head impaction
- ❑ Critical nerves at risk: Axillary nerve (15-20% of cases)
- ❑ Main **Static Stabilizer factors**
 - Glenoid labrum: Deepens socket by 50%
 - Inferior glenohumeral ligament (IGHL)
- ❑ **Dynamic Stabilizers factors**
 - Rotator cuff:
 - Long head of biceps



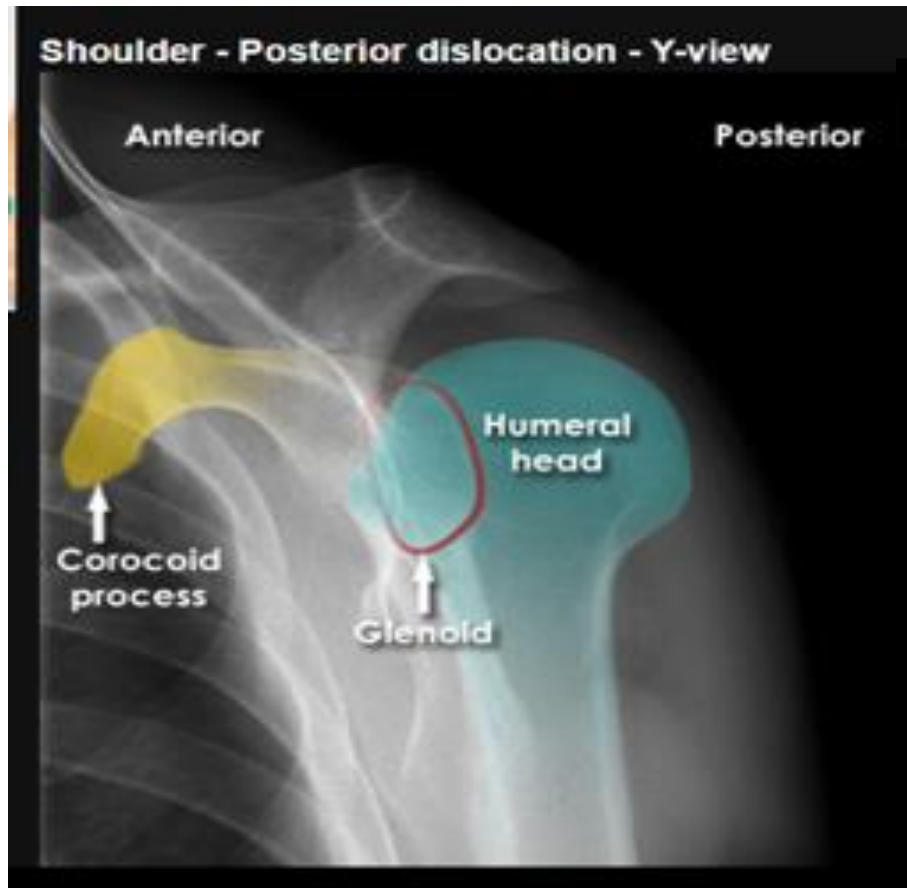
Posterior dislocation of shoulder dislocation

Case scenario: A 45-year-old man presents to the ER after an **electrocution** injury. He complains of severe shoulder pain and inability to lift his arm. On exam, his arm is fixed in **internal rotation**, and he has limited **external rotation**.



Posterior dislocation of shoulder dislocation

- ❑ **Mechanism:** Seizure or electric shock (violent muscle contractions).
- ❑ **Often missed!** Up to 50% are overlooked on initial presentation.



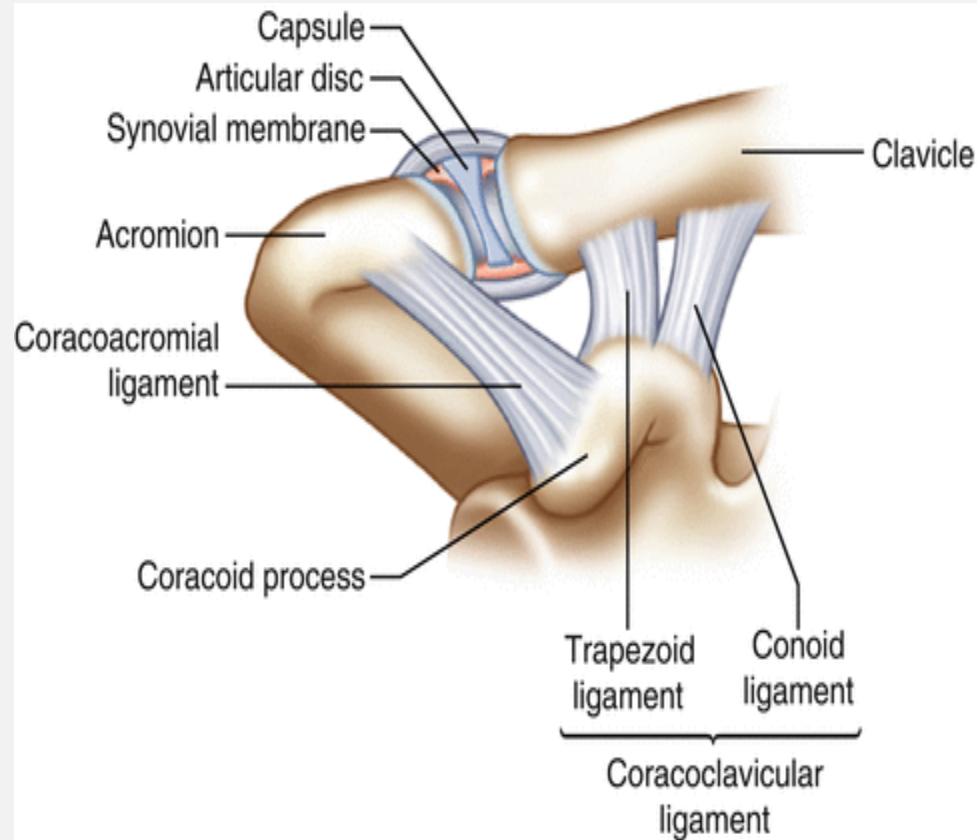
Posterior dislocation of shoulder dislocation

- ❑ **Mechanism:** Seizure or electric shock (violent muscle contractions).
- ❑ **Often missed! Up to 50% are overlooked on initial presentation.**
- ❑ **Affected Structures:**
 1. Posterior labrum: labral tear
 2. Teres minor and infraspinatus: May be injured due to forceful internal rotation.
- ❑ **Glenoid is naturally retroverted**, making posterior dislocation less common but high-risk for chronic instability if missed.



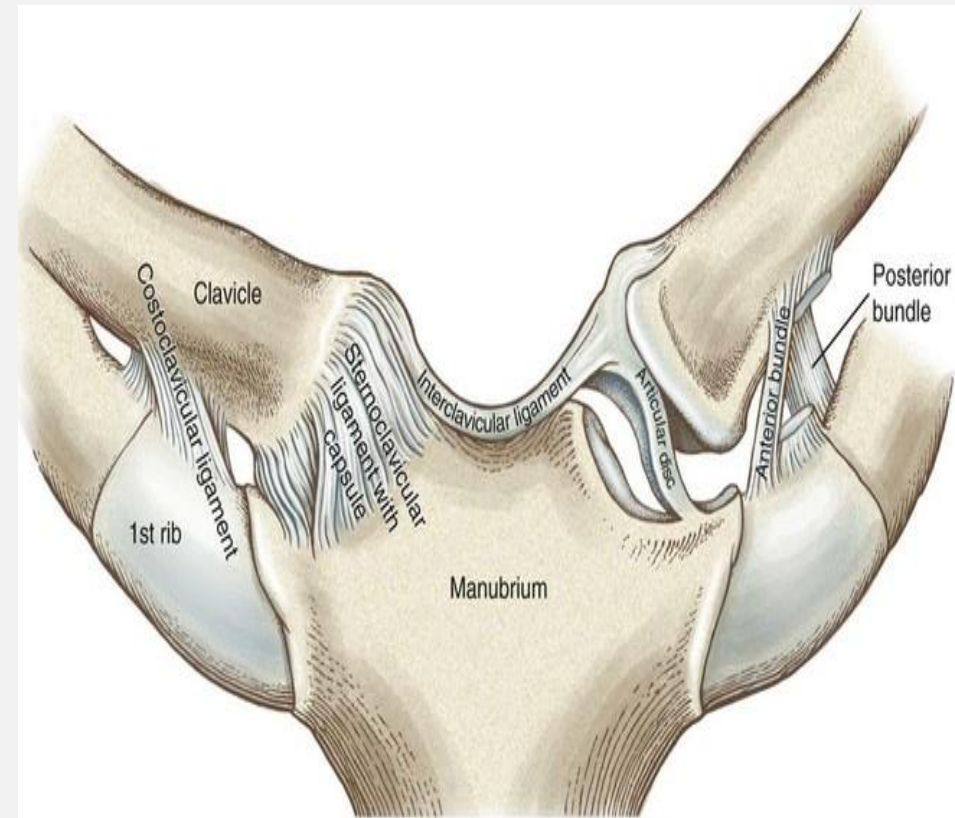
Acromioclavicular (AC) Sternoclavicular (SC) Joints Injuries

AC joint



- ❑ Type: Synovial, plane variety
- ❑ Stability . Coraco-clavicular ligament
- ❑ Intra-articular disc Incomplete
- ❑ Function: Shares a little in movements of shoulder girdle

SC joint



- ❑ Type: Synovial joint ; modified saddle variety
- ❑ Stability . Costo-clavicular ligament
- ❑ Intra-articular disc complete
- ❑ Function: main joint for movements of shoulder girdle

MOVEMENT ON THE SHOULDER GIRDLE

A. Elevation

1. Upper fibres of trapezius
2. Levator scapulae

B. Depression

1. Pectoralis minor
2. Pectoralis major

C. Rotation up (glenoid cavity faces upwards):

1. Upper and lower fibres of trapezius
2. Serratus anterior

D. Rotation down (glenoid cavity faces downwards):

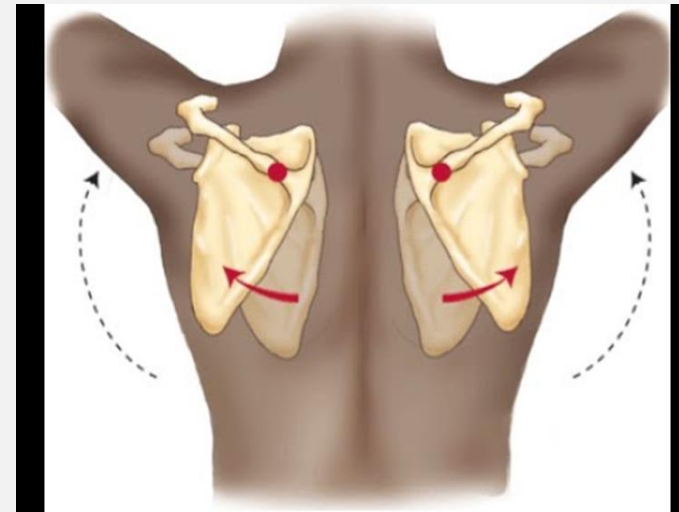
1. Levator scapulae
2. Rhomboids major
3. Rhomboids minor

E. Protraction (pulling forwards):

- 1. Serratus anterior
- 2. Pectoralis minor

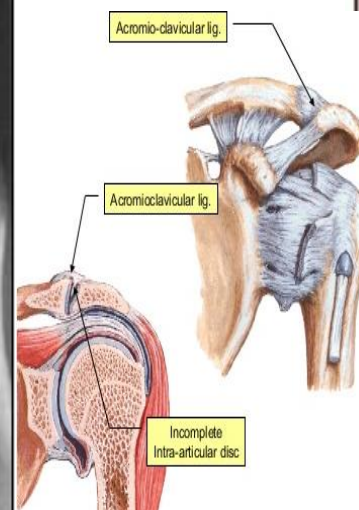
F. Retraction (pulling backwards towards the vertebrae):

- 1. Middle fibres of trapezius
- 2. Rhomboids major
- 3. Rhomboids minor



Acromioclavicular (AC) Joint Injury

Case scenario: A 25-year-old cyclist falls directly onto his right shoulder during a race. He presents with **localized pain over the AC joint**, **a visible step-off deformity**, and **pain with cross-body adduction**.



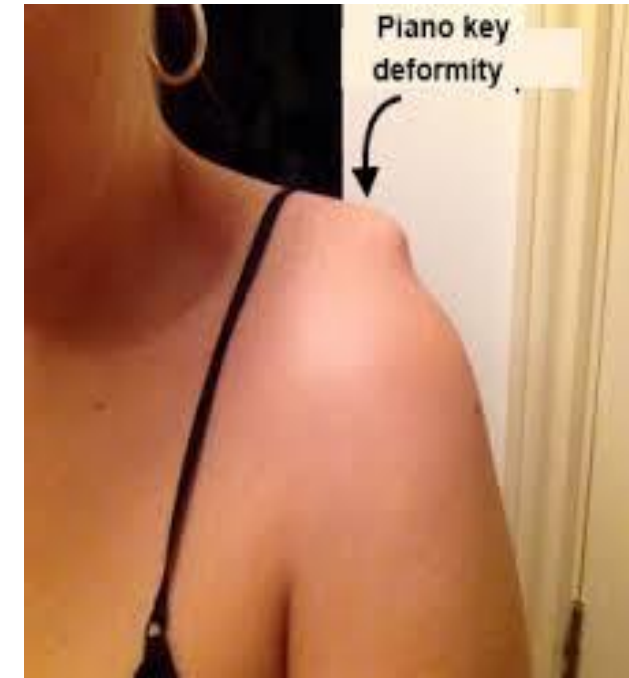
Acromioclavicular (AC) Joint Injury

Key Features:

- Mechanism: Direct trauma (fall onto shoulder) or indirect (fall on outstretched hand).
- Classic Sign: **"Piano Key" sign** (AC joint reducible with downward pressure).
- High-Risk Groups: Contact athletes (hockey), cyclists, military personnel.

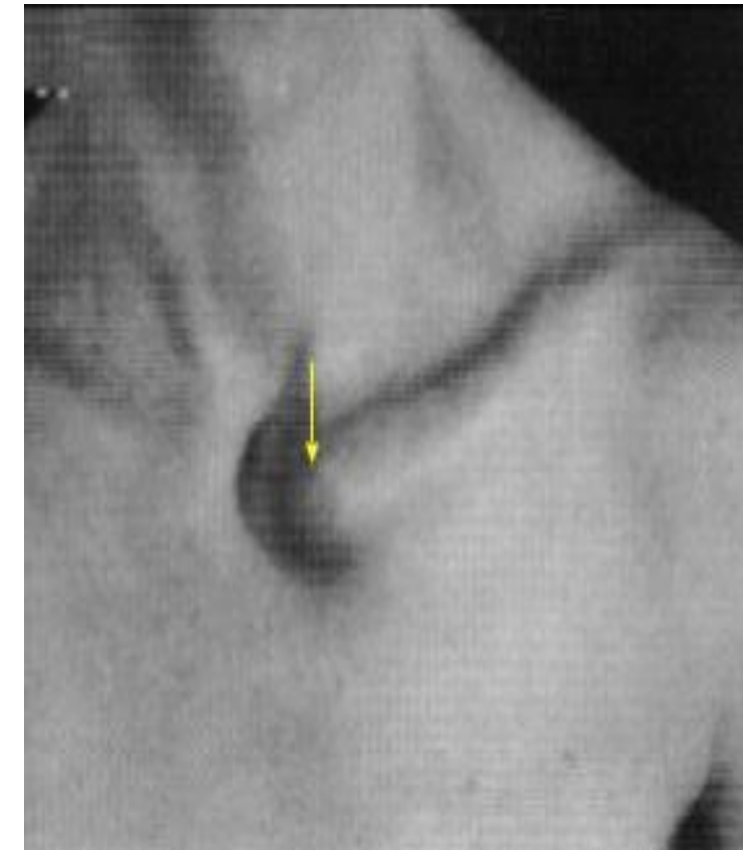
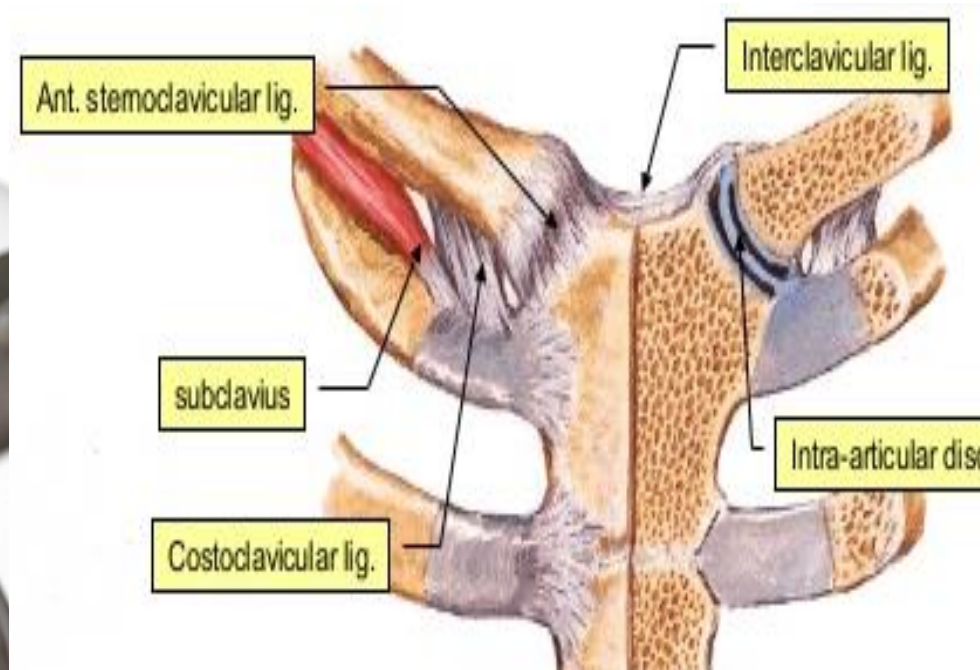
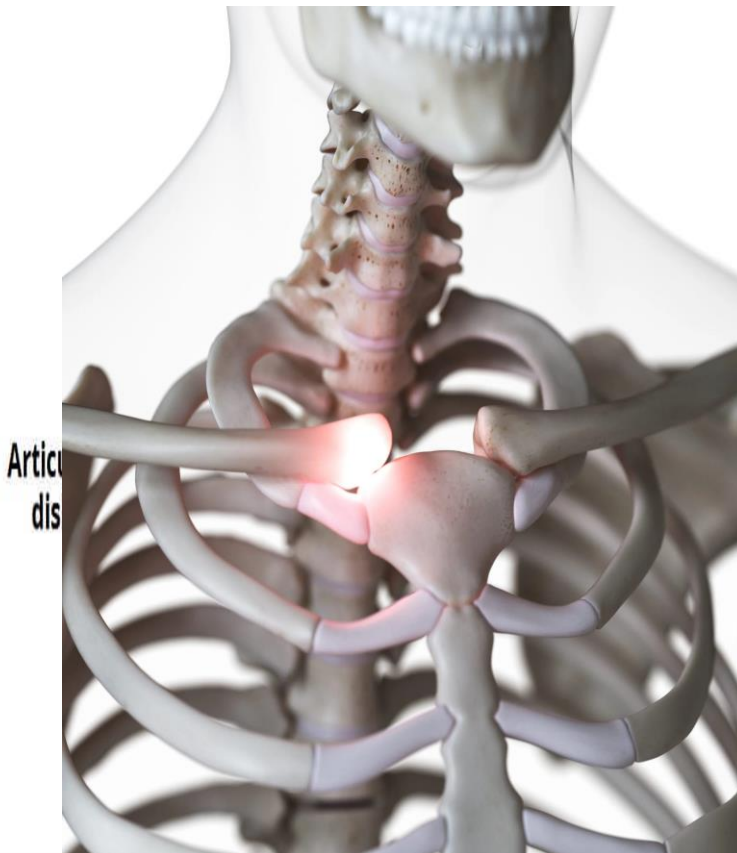
Affected Structures:

- Coracoclavicular (CC) Ligaments (**Conoid + Trapezoid**): Primary vertical stabilizers.
- AC Ligaments: Primary horizontal stabilizers.
- **CC ligaments** prevent superior clavicular displacement; **injury → step deformity**



Sternoclavicular Joint Injuries

Case scenario: 20-year-old player sustains a direct impact to the front of his chest during a training. He presents with **localized pain** and **swelling over the medial clavicle**, **worsening with arm movement**.



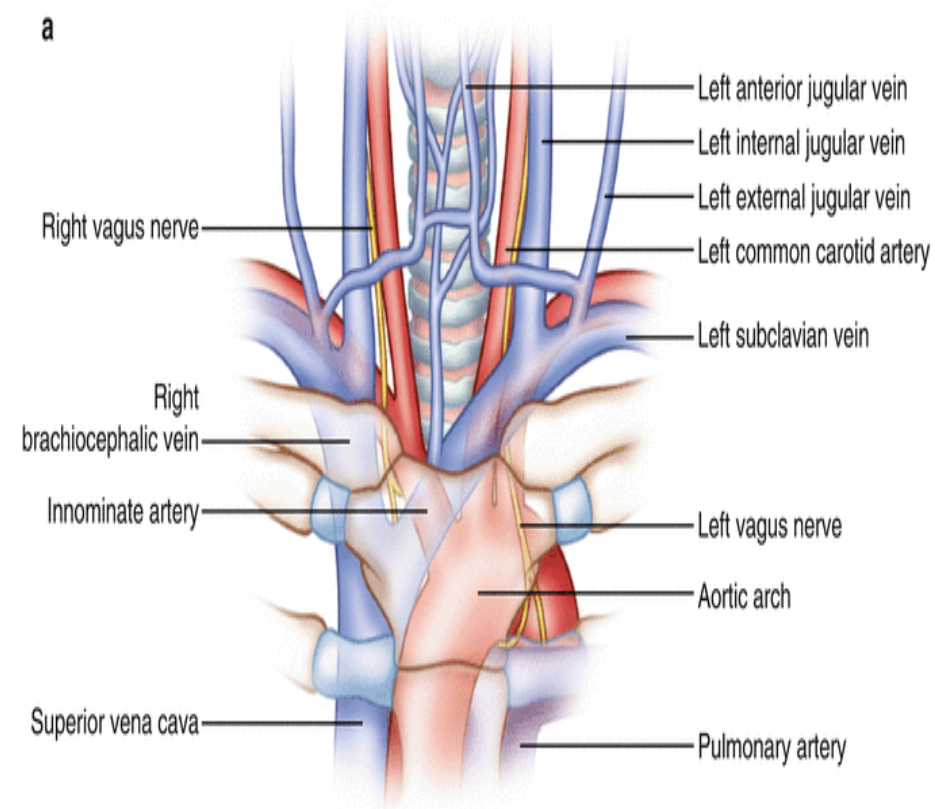
Acromioclavicular (AC) Joint Injury

Key Features:

- Mechanism: Direct trauma (e.g., impact) or indirect (fall on outstretched arm).
- Red Flags: Dyspnea/dysphagia (mediastinal compression).
- Anatomical Uniqueness: Only synovial joint connecting axial to appendicular skeleton..

Anatomical features:

- **Articular surfaces:** Sternal clavicle + manubrium + 1st rib cartilage.
- **Stabilizers:** **Ant./Post. SC ligaments** (prevent ant./post. displacement). **Costoclavicular ligament** (resists superior displacement). **Interclavicular ligament** (connects medial clavicles).
- **Vital Posterior Relations:** Brachiocephalic vein, subclavian artery, trachea/esophagus, pleura.
- **Posterior dislocations** risk life-threatening mediastinal injury (vascular/airway compromise).
- **Anterior dislocations** more common but less dangerous.



Clavicular fractures

Clavicular fractures

Case scenario: A 15-year-old boy falls off his bike onto his left shoulder. He presents holding his **arm adducted and supported at the elbow**, with visible **deformity and swelling over the clavicle**



Clavicular fractures

Key Features:

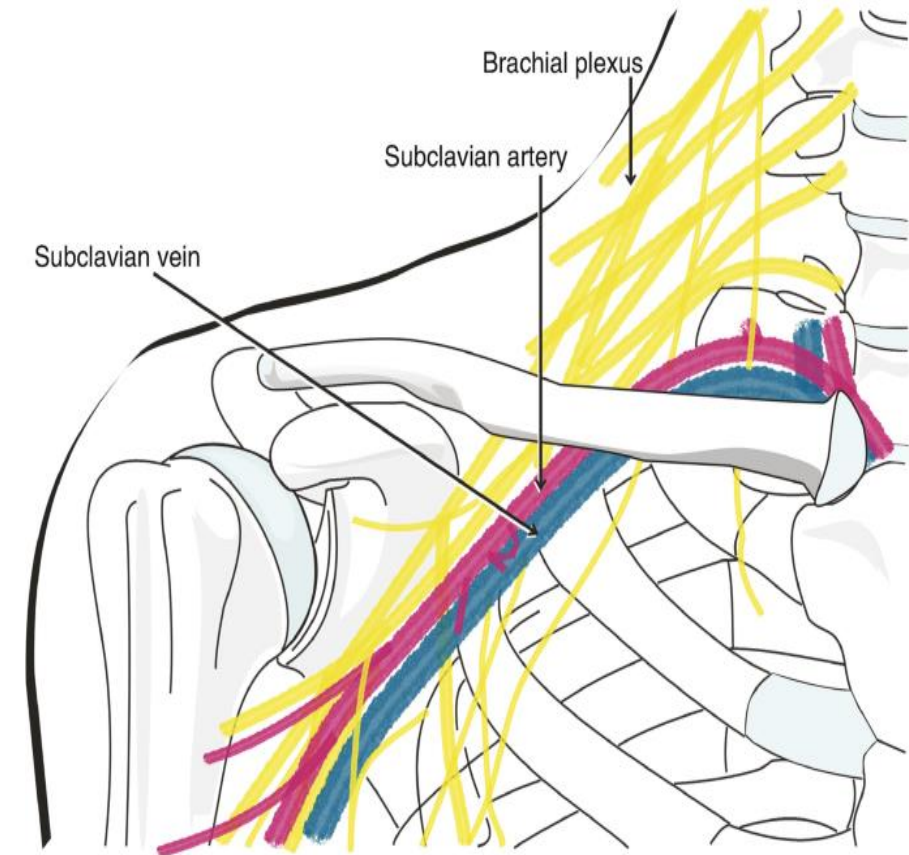
- **Most Common Mechanism:** Fall onto shoulder (85%) or direct trauma.
- **Classic Sign:** "Sternal droop" (shoulder slumps downward due to loss of clavicular strut).
- **High-Risk Groups:** Adolescents (midshaft fractures), newborns (birth trauma), contact athletes...

Types of fractures:

- Medial 1/3: Least common (5%), high risk for mediastinal injury.
- **Middle 1/3: Most common (80%),** subcutaneous and vulnerable.
- Lateral 1/3: High nonunion risk due to coracoclavicular ligament disruption.

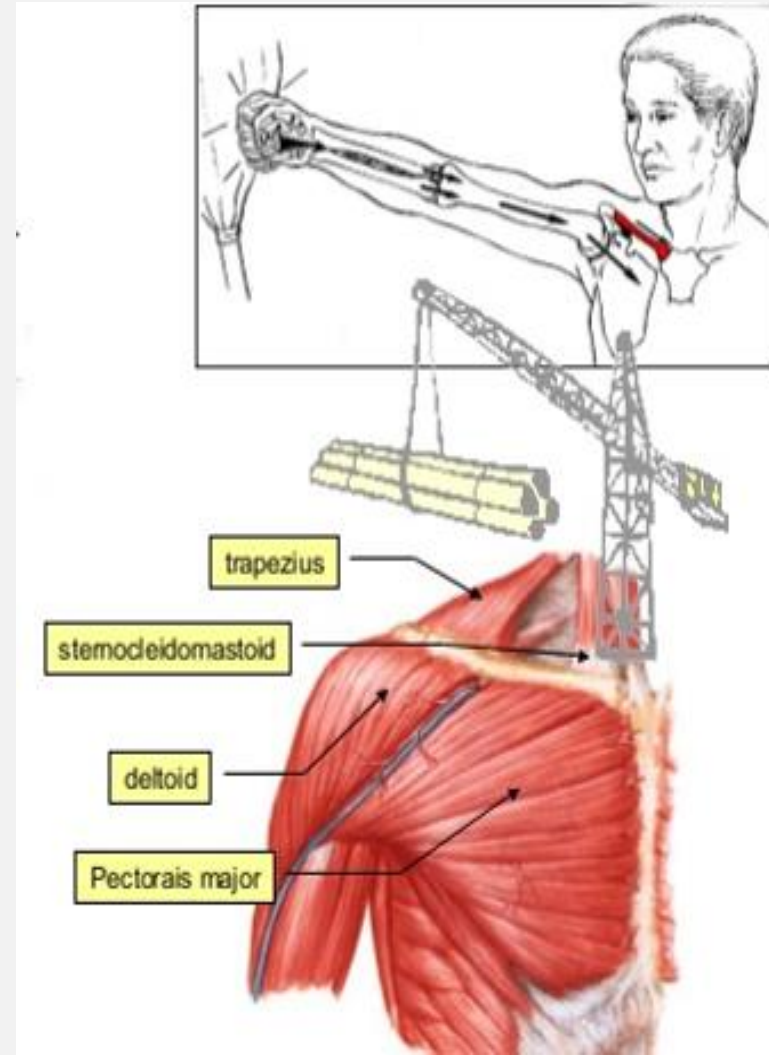
Critical Adjacent Structures:

- Subclavian vessels & brachial plexus (posterolateral to medial clavicle).
- Pleura/lung (risk of pneumothorax).



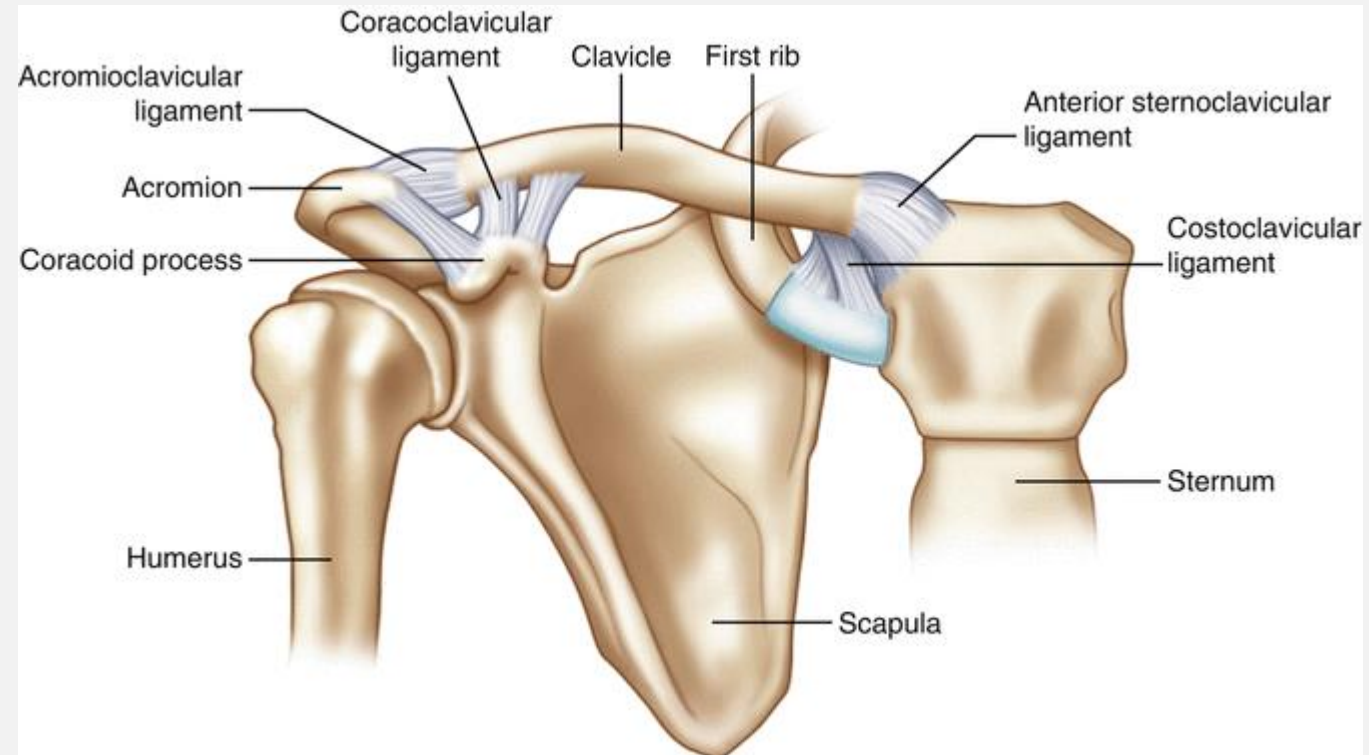
❑ what is the function of clavicle?

- Transmitted forces from upper limb to axial skeleton
- Acts as a strut holding the arm free from trunk to hang supported principally by trapezius
- Provides attachment to muscles



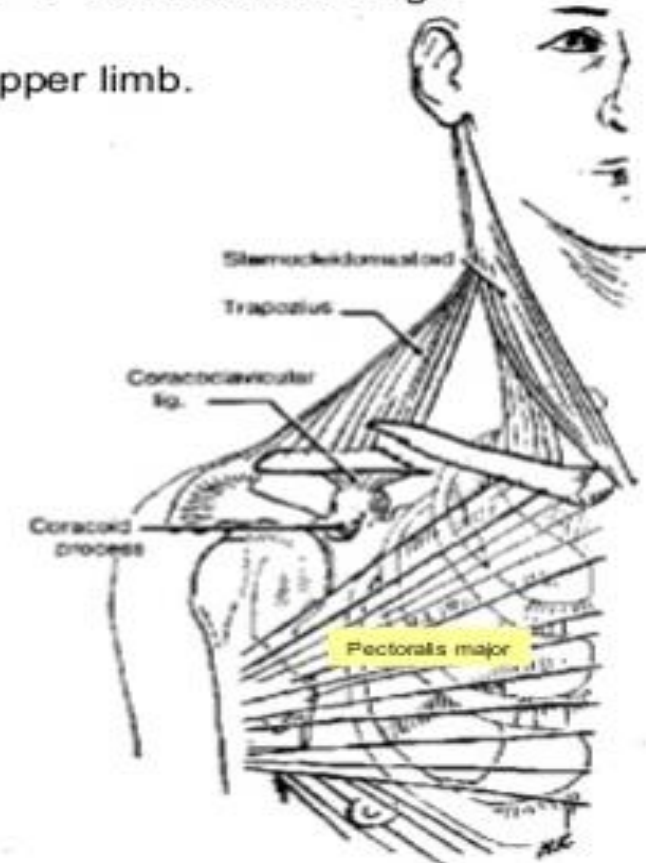
□ why fracture of the clavicle is common?

- Almost subcutaneous
- Relatively fixed



❑ what is the commonest site for fracture in clavicle ? Explain why fracture segments are displaced

- **Common site:** at the junction of the middle and lateral thirds (weakest point)
- **Anatomy of displacement:**
 - Trapezius is unable to support the weight of the arm → The shoulder sags slightly owing to the weight of the limb.
 - Pectoralis major adducts and medially rotates the upper limb.
 - Sternocleidomastoid elevates the inner fragment.



❑ why the commonest sit for fracture clavicle is at junction between medial 2/3 with lateral 1/3?

because it is the thinnest point in the clavicle as it

- 1) Point between 2 different curved areas
- 2) Point between 2 different cross-section areas
- 3) Weakened by presence of subclavius groove in its inferior surface
- 4) Weakened by presence of nutrient foramen for entrance of nutrient artery
- 5) Unprotected as it has no muscle attachment

Degenerative diseases of shoulder joint

Frozen Shoulder (Adhesive Capsulitis)

Case scenario: A 55-year-old **diabetic man** presents with gradual onset **shoulder pain and stiffness** over 6 months. She reports **pain at night**, difficulty dressing, and **inability to reach behind her back**. **Passive and active range of** motion (ROM) are globally restricted, especially in **external rotation and abduction**.



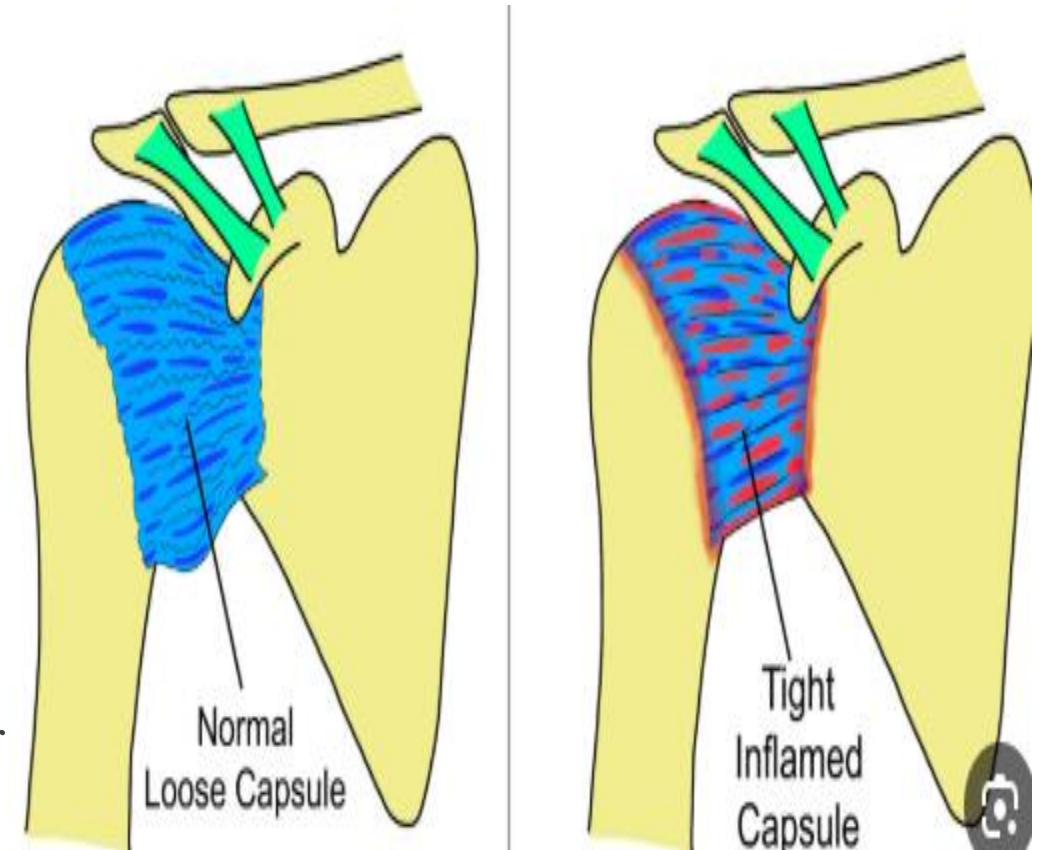
Frozen Shoulder (Adhesive Capsulitis)

Key Features:

- Classic Triad: Pain + Stiffness + Loss of ROM (active and passive).
- High-Risk Groups: Diabetics (4x risk), females aged 40–60, thyroid disorders

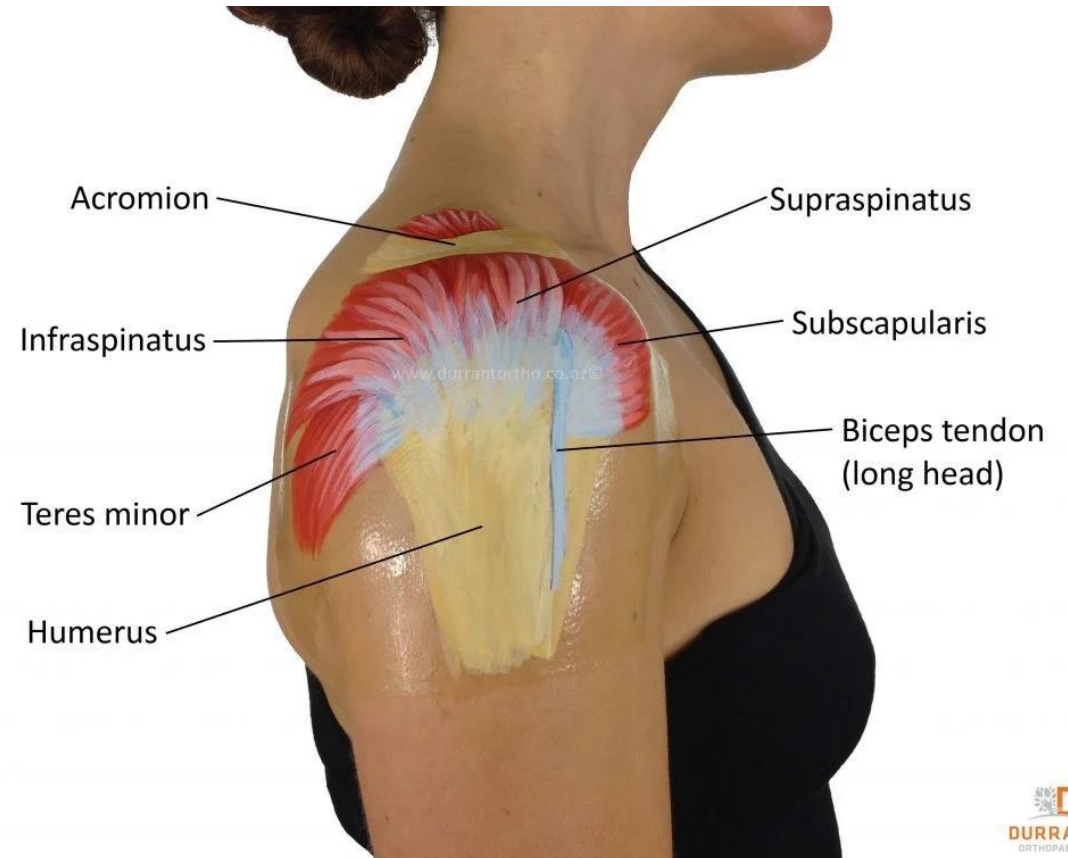
Affected Structures:

- **Glenohumeral Joint Capsule:** Thickened and contracted (especially coracohumeral ligament).
- **Synovium:** Inflamed in early stages.
- **Axillary Recess:** Obliterated, limiting abduction.
- Loss of External Rotation = **Pathognomonic** (due to coracohumeral ligament contracture).
- No Primary Joint Damage: **Unlike osteoarthritis**, the articular cartilage remains intact.



Rotator Cuff Tears

Case scenario: A 60-year-old woman presents with gradual right shoulder pain worsening over 6 months. She reports night pain, weakness when lifting objects, and difficulty combing her hair. On exam, she has positive "drop arm sign" and weakness in abduction/external rotation.



Rotator Cuff Tears

Key Features:

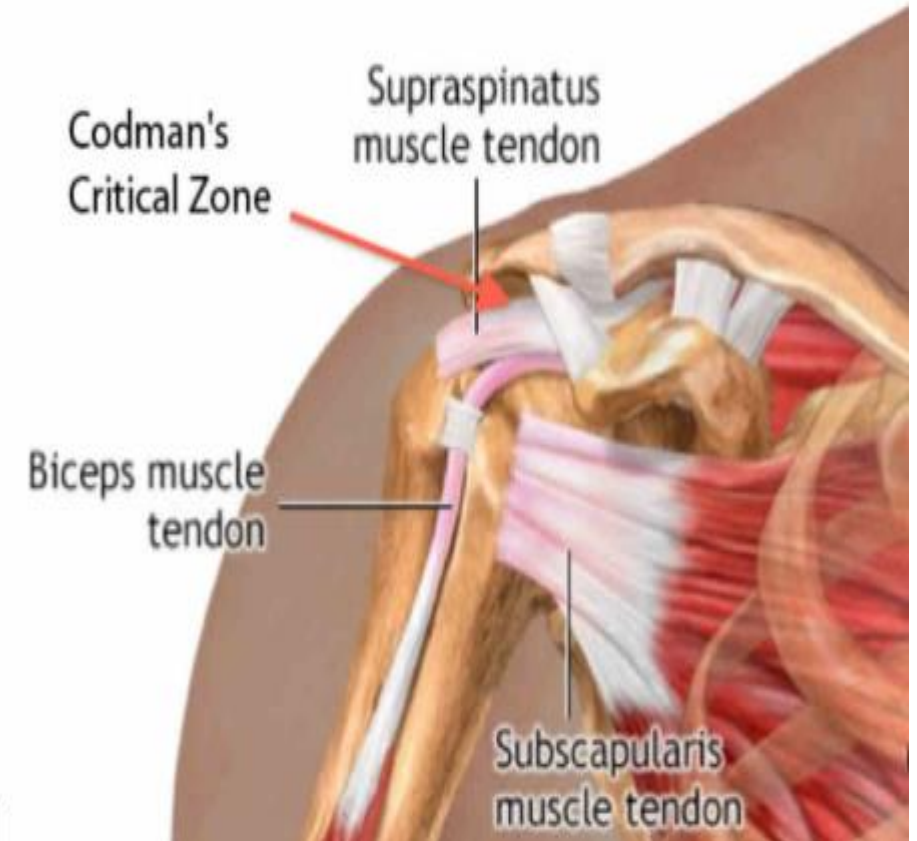
- **Most common cause:** Degenerative changes (age >40) or trauma (e.g., fall on an outstretched arm).
- **Classic triad:** Pain + Weakness + Limited active ROM (passive ROM preserved).
- **High-risk groups:** Manual laborers, overhead athletes (e.g., baseball pitchers).

Affected Structures:

- **Supraspinatus tendon** (most commonly torn).
- **Infraspinatus/teres minor** (external rotation weakness).
- **Subscapularis** (rare isolated tears; affects internal rotation).
- **Coracoacromial arch** (impingement contributor).

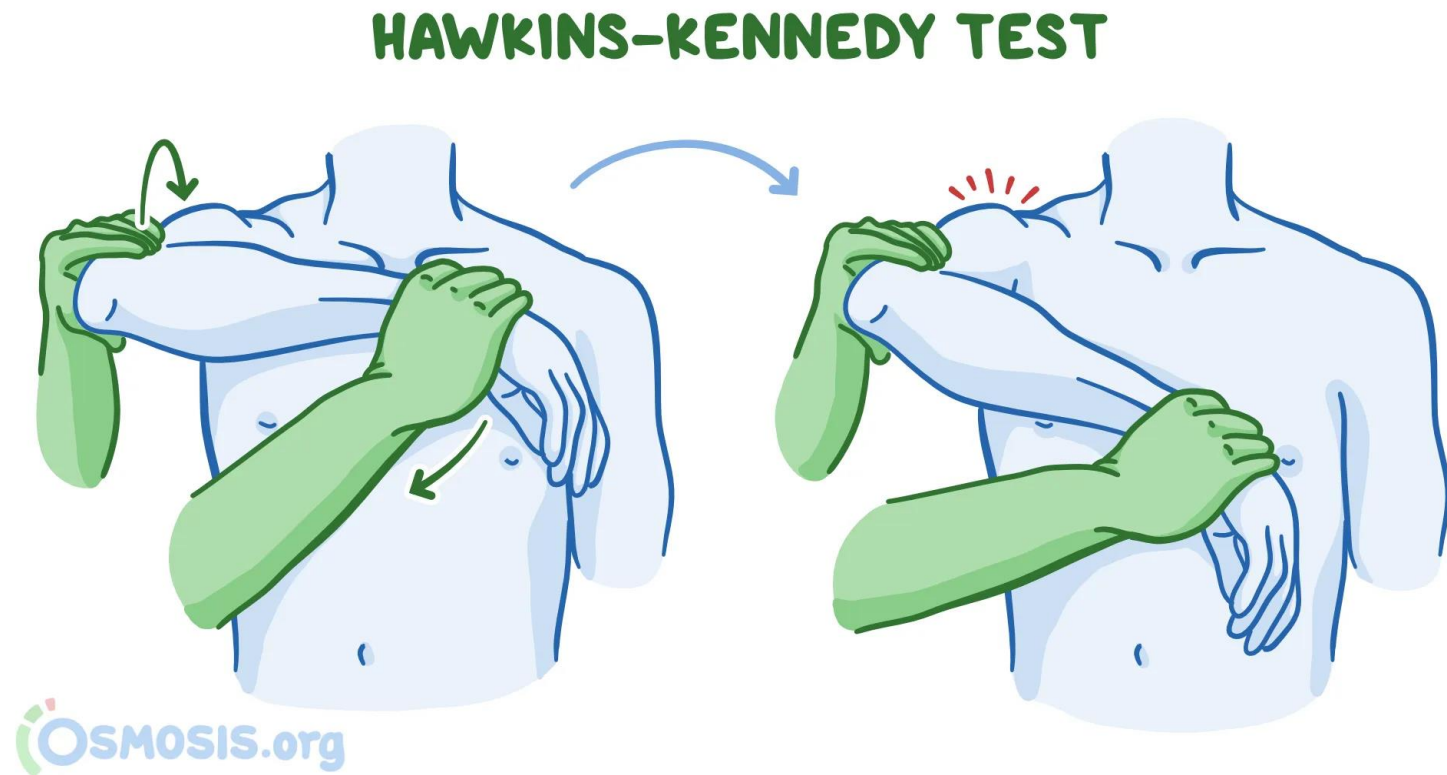
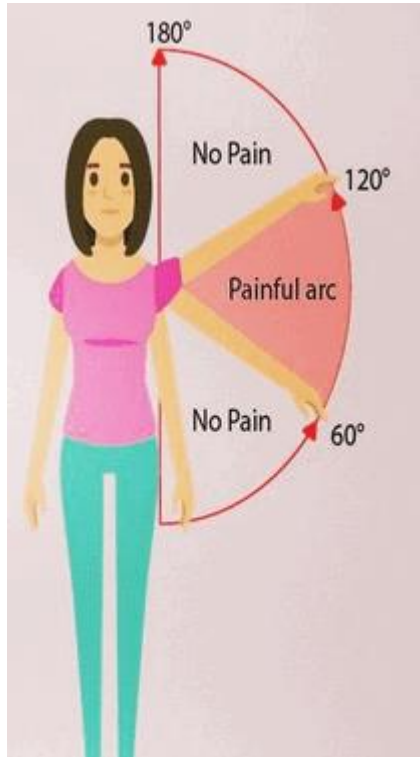
What happen

- thickening of the cuff fibers "**rotator cable**"
- Critical zone: Hypovascular area near supraspinatus insertion → prone to degeneration.



Shoulder Impingement Syndrome

Case scenario: A 45-year-old construction worker presents with progressive shoulder pain when lifting objects overhead. Symptoms **worsen at night**, especially when lying on the affected side. Examination reveals **painful arc (60°-120° abduction)**, **weakness in supraspinatus testing**, and Hawkins tests..



Shoulder Impingement Syndrome

Key Features:

- **Classic Triad:** Painful arc + Night pain + Weakness with overhead activities.
- **Mechanism:** Compression of rotator cuff tendons and subacromial bursa under the coracoacromial arch.
- **High-Risk Groups:** Overhead athletes (swimmers), manual laborers, diabetics.

Affected Structures:

1- Rotator Cuff Tendons:

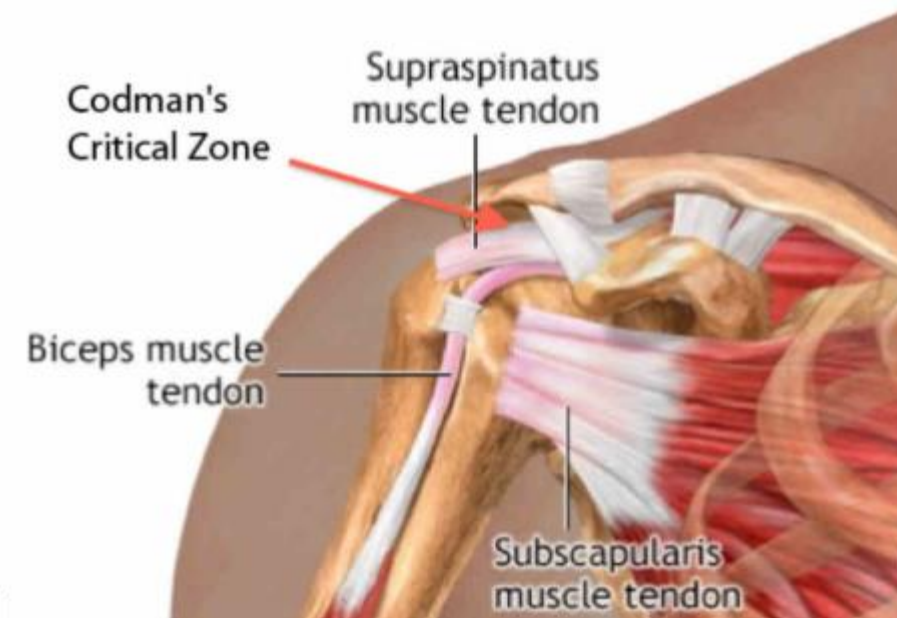
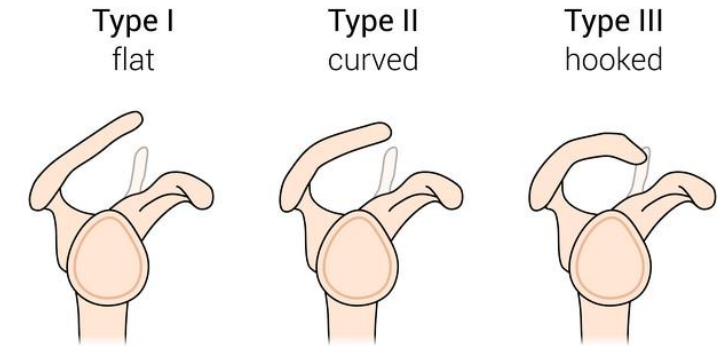
- Supraspinatus (most commonly impinged).
- Infraspinatus/Subscapularis (later stages).

2- Subacromial Bursa: Inflamed, causing "bursitis."

3- Coracoacromial Arch:

- Acromion types (Bigliani classification): Flat (I), Curved (II), Hooked (III).
- Hooked (III) acromion ↑ impingement risk.

Acromial morphology



Brachial Plexus Injuries

Brachial Plexus Injuries

Case scenario: After a difficult forceps delivery of a male infant, the baby presents later with his left arm was medially rotated and adducted, while the forearm was pronated and extended



Brachial Plexus branches

Branches of brachial plexus are:

From the roots:

- Dorsal scapular nerve (C5)
- Long thoracic nerve (C5, C6, C7)

From the upper trunk:

- Suprascapular nerve (C5, C6)
- Nerve to subclavius (C5, C6)

From the three cords of brachial plexus.

▪ Branches of lateral cord – Three

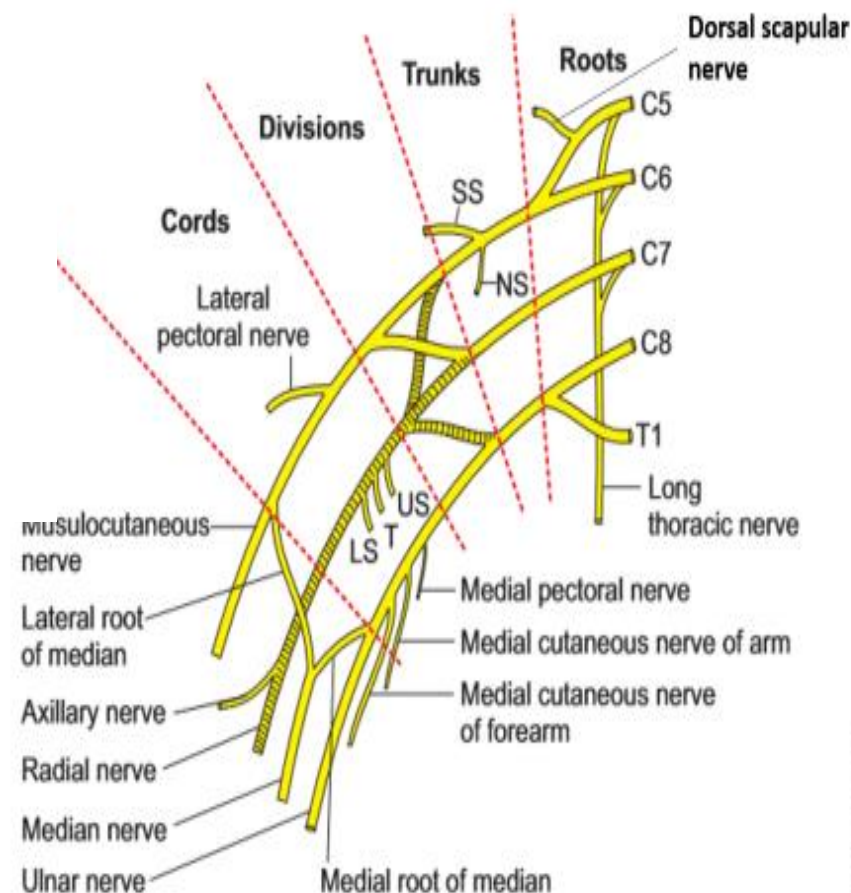
- Lateral pectoral nerve (C5-7)
- Lateral root of median nerve (C5-7)
- Musculocutaneous nerve (C5-7)

▪ Branches of medial cord – Five

- Medial pectoral nerve (C8, T1)
- Medial cutaneous nerve of arm (C8, T1)
- Medial cutaneous nerve of forearm (C8, T1)
- Medial root of median nerve (C8, T1)
- Ulnar nerve (C8, T1)

Branches of posterior cord – Five

- Upper subscapular nerve (C5, C6)
- Lower subscapular nerve (C5, C6)
- Thoracodorsal nerve (C6-8)
- Axillary nerve (C5, C6)
- Radial nerve (C5-T1)



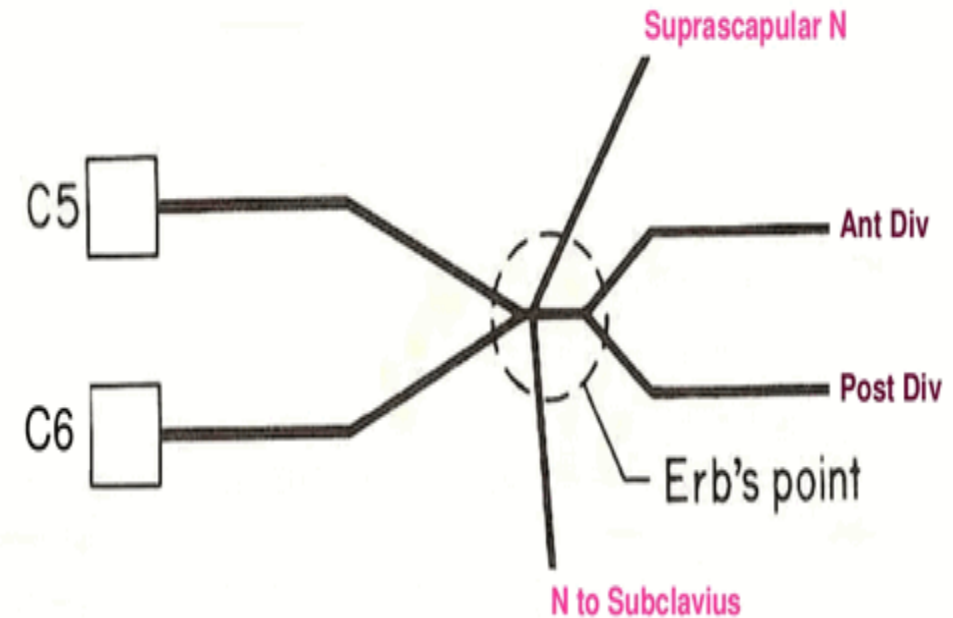
SS- suprascapular nerve
NS- Nerve to subclavius
US – upper subscapular N.
T – Thoracodorsal N.
LS- lower subscapular nerve

Brachial Plexus branches

□ Erb's point in brachial plexus

meeting point of six nerves , which includes

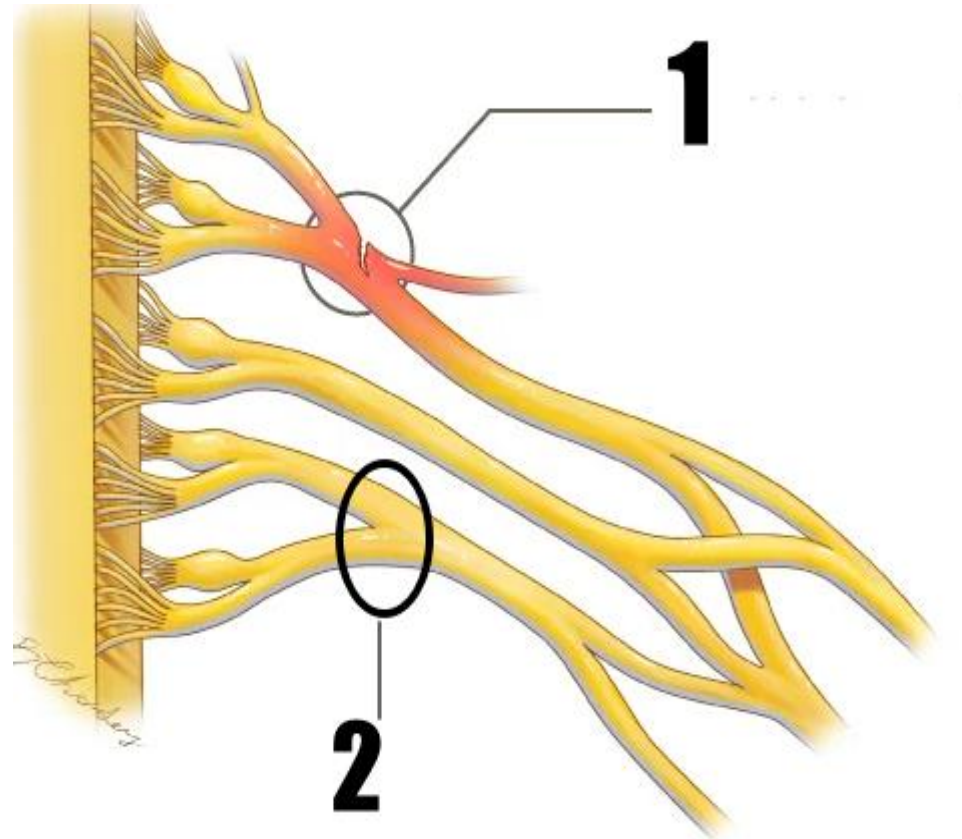
- ventral rami of C5 and C6
- two branches (suprascapular and nerve to subclavius) of upper trunk
- two divisions of upper trunk



Brachial Plexus Injuries

Key Anatomical Clues:

- ❑ **Upper trunk injury** (Erb's palsy):
- ❑ **Lower trunk injury** (Klumpke's palsy):



Brachial Plexus Injuries

❑ Upper trunk injury (Erb's Douchenne paralysis)

- damage to the **CS and C6** fibers of the superior trunk
- **Causes:**
stretching or tearing the upper brachial plexus in the posterior triangle region of the neck as in
 - a) Falls on the shoulder in which head & shoulder are separated
 - b) Abstracted labour

Nerves affected:

Musculocutaneous, axillary, suprascapular, and N. subclavius

Motor deficits:

- 1) Paralysis of abductor of shoulder, flexor of elbow and supinator muscles leading to combined flat shoulder and **policeman tip deformity** (adduction of arm, extension & pronation of forearm).
- 2) Wasting of scapular muscles, front of arm with flat shoulder.

Sensory deficit:

- Sensory loss over lower 1/2 of deltoid, lateral aspect of arm, lateral aspect of forearm, lateral aspect of hand & lateral 2 fingers



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Sensory deficit:

- Sensory loss over lower 1/2 of deltoid, lateral aspect of arm, lateral aspect of forearm, lateral aspect of hand & lateral 2 fingers



❑ Lower trunk injury (Klumpke's palsy)

damage to the C8 and T1 fibers of the lower trunk

Causes:

- occurs when upper limb is suddenly pulled upward and away from the body.
- Obstetric injury (shoulder dystocia)

Nerves affected:

C8 and T1 lesions primarily affect

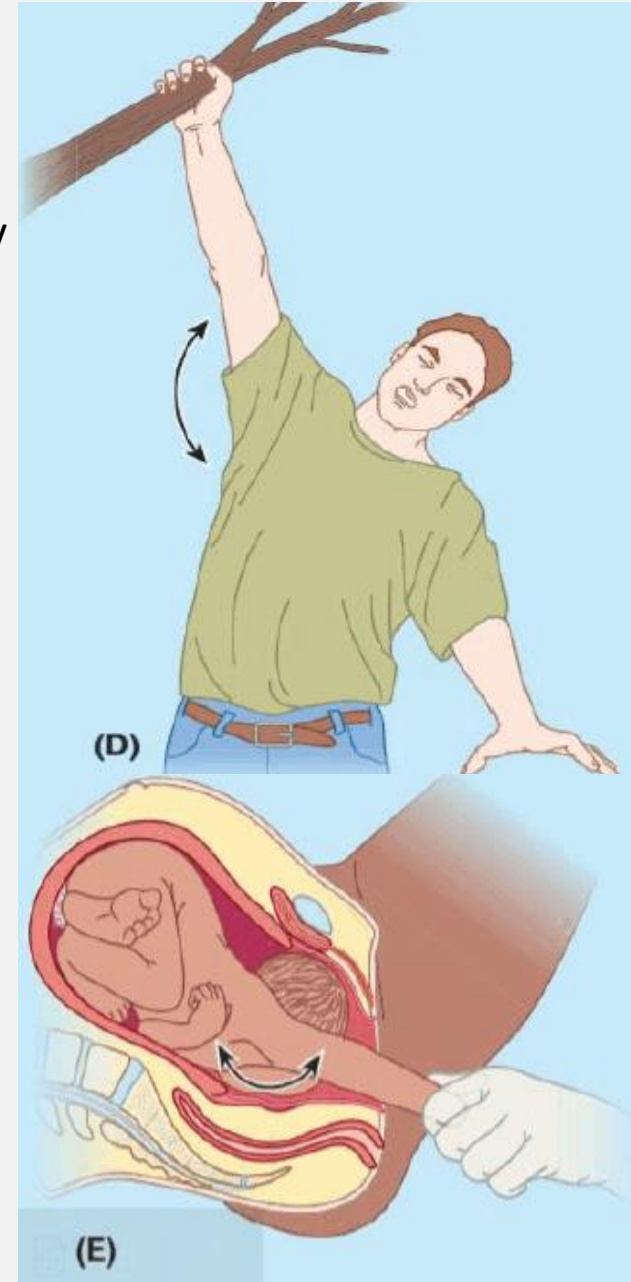
- 1) intrinsic muscles of hand innervated by ulnar nerve (**interossei & medial two lumbricals**)
- 2) some of muscles innervated by C8 and T1 fibers of median nerve (**thenar compartment**)

Motor deficits:

- 1) Patient presents with a **combined claw** hand and some degree of ape hand.
- 2) In addition, there is some weakness of the thenar muscles.

Sensory deficit:

- Sensory loss of the medial border of the forearm and the medial 1 & 1/2 digits



❑ Lower trunk injury (Klumpke's palsy)

damage to the C8 and T1 fibers of the lower trunk

Causes:

- occurs when upper limb is suddenly pulled upward and away from the body.
- Obstetric injury (shoulder dystocia)

Nerves affected:

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Motor deficits:

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- 2) In addition, there is some weakness of the thenar muscles.

Sensory deficit:

- Sensory loss of the medial border of the forearm and the medial 1 & 1/2 digits



Paralysis of all small muscles of hand (C8&T1)

median & Ulnar nerves

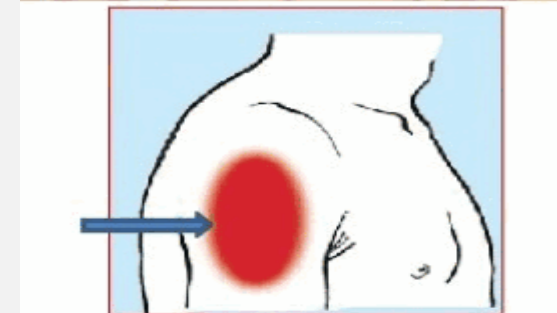
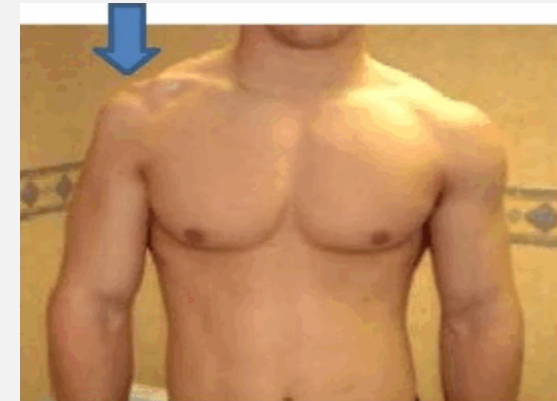
- Hyperextended MCP joints
- Flexion of IP joints

Injuries of branches

what is the effect of axillary nerve injury?

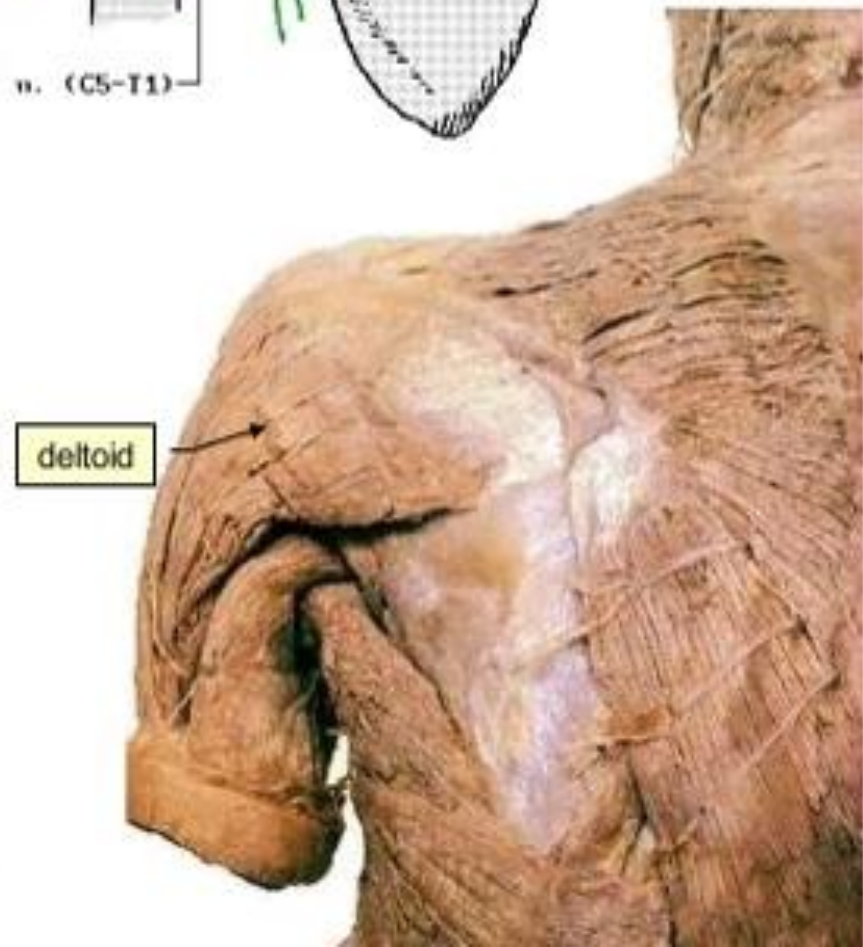
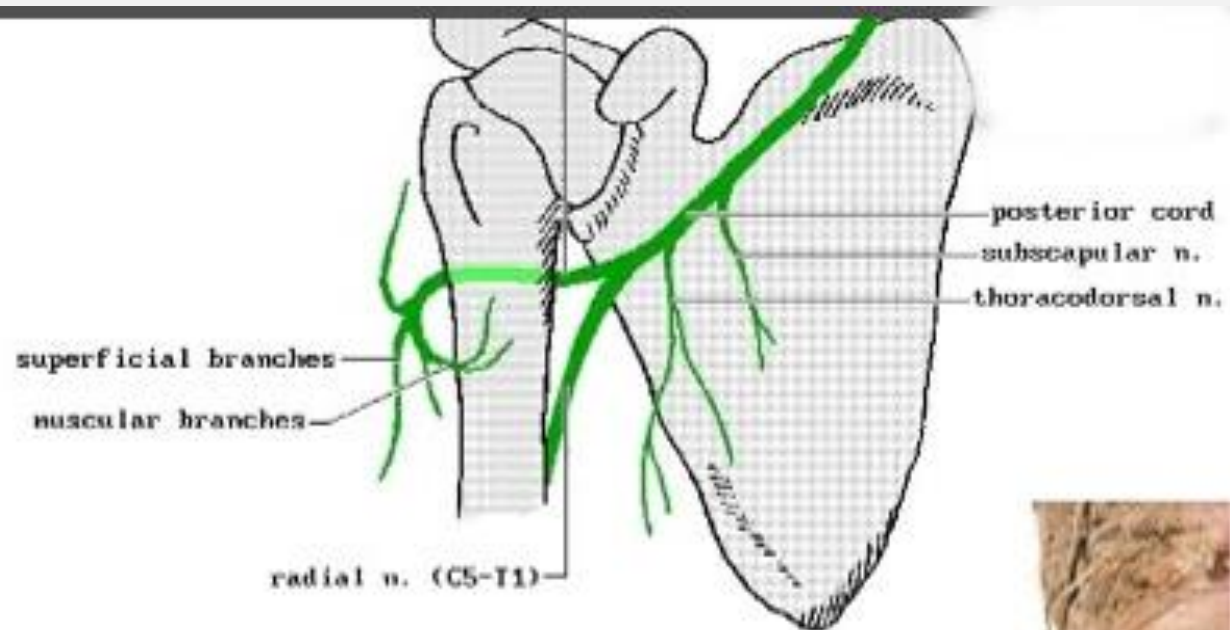
- **Root :** (C5-C6)
- **Common lesion sites:**
 - (1) fracture of the surgical neck of the humerus
 - (2) inferior dislocation of the shoulder joint.
- **Muscles innervated:** **deltoid & teres minor muscles.**
- **Motor deficits:**
 - 1) Loss of horizontal abduction (15- 110 degrees) of the arm
 - 2) Atrophy of the deltoid muscle and loss of the rounded contour of the shoulder.
 - 1) Weak lateral rotation of arm
- **Sensory deficit:**

Sensory loss of skin overlying the cap of the shoulder..



Axillary nerve

- **Origin and branches:**
 - Muscular: deltoid and teres minor.
 - Sensory: upper lateral cutaneous nerve of the arm.
- **Testing.**



❏ what is the effect of Suprascapular nerve injury?

- **Root :** (C5-C6)

- **Common lesion sites:**

Injury due to entrapment as the nerve crosses the superior border of the scapula in the scapular notch

Muscles innervated:

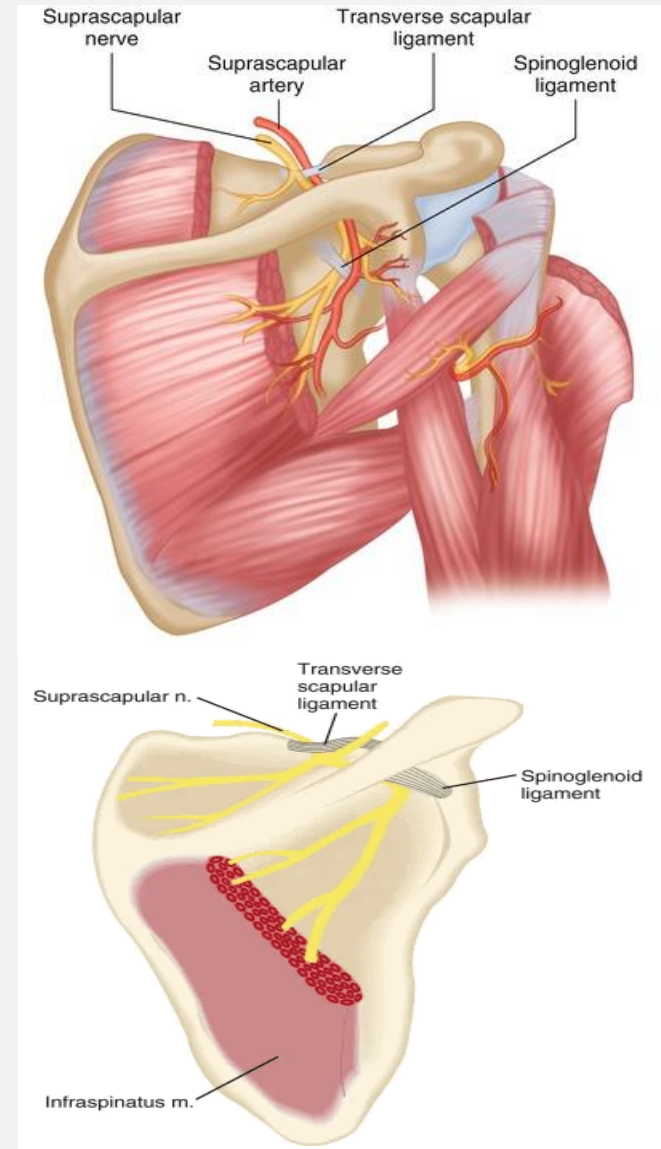
supraspinatus & Infraspinatus.

- **Motor deficits:**

- 1) Weakness in initiating the first 15 degrees of shoulder abduction
- 2) Some weakness of external rotation of the shoulder.

Sensory deficit:

- No sensory loss



❑ what is the effect of Long Thoracic Nerve injury?

- **Root :** (C5,C6 , C7)
- **Common lesion sites:**

Damaged as the nerve passes on the superficial surface of the Serratus anterior muscle by trauma to the lateral chest wall or following **mastectomy** when the nerve may be accidentally cut at the lateral aspect of the mammary gland

Muscles innervated:

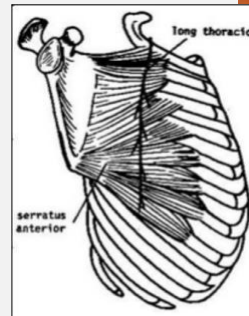
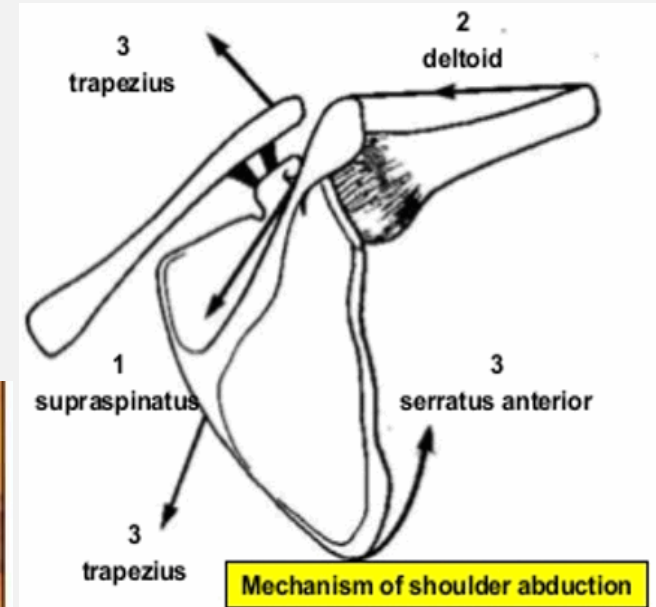
Serratus anterior muscle .

▪ Motor deficits:

- 1) difficulty in overhead abduction
- 2) Weakness protraction
- 3) Sign is a winged scapula

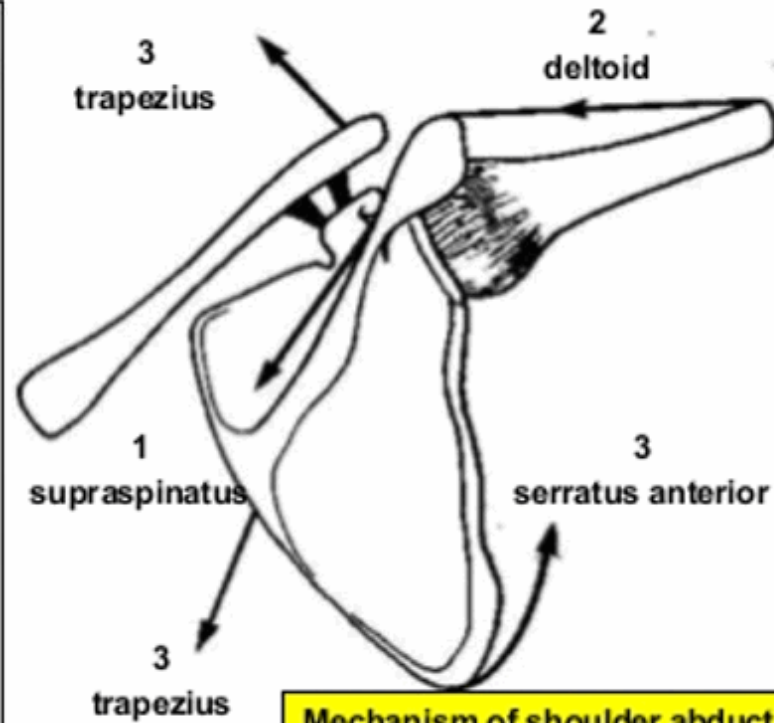
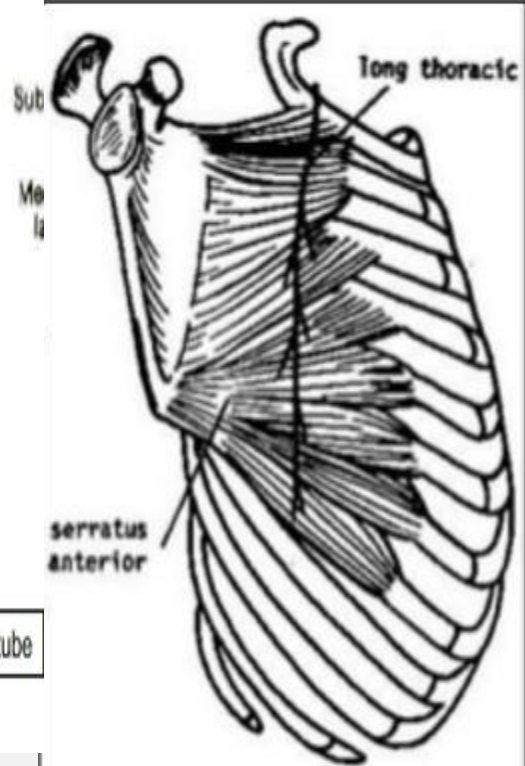
Sensory deficit:

- No sensory loss





Chest tube



❑ what is the effect of Thoracodorsal nerve injury?

- **Root :** (C6-C7-C8)
- **Common lesion sites:**
Injuries occur with trauma or surgical approaches to the lateral chest wall

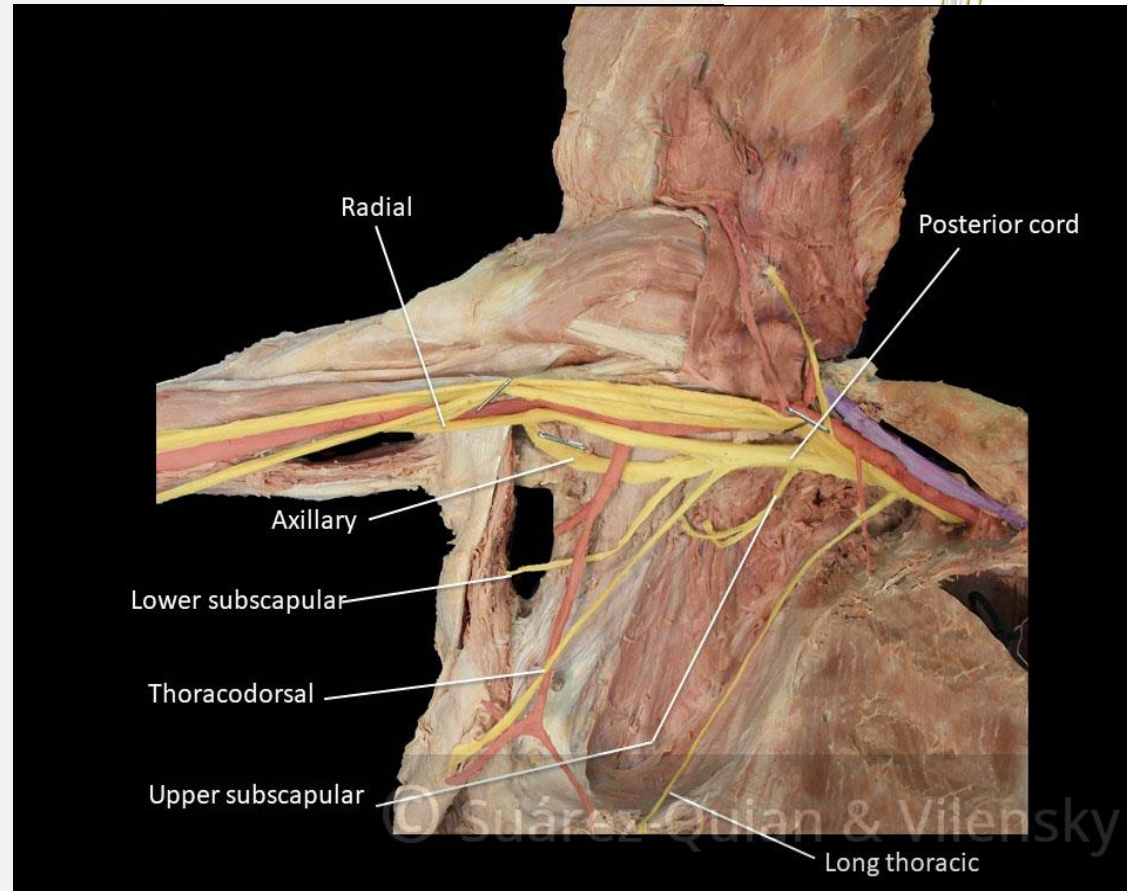
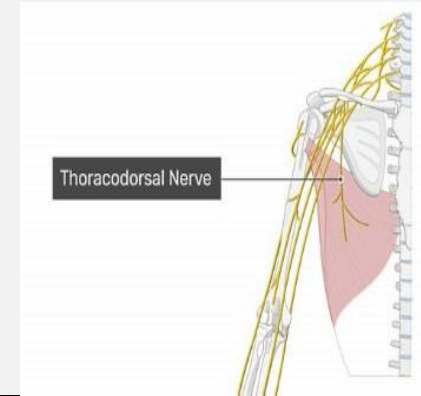
Muscles innervated:

lattissimus dorsi muscle

- **Motor deficits:**
1) Weakened adduction, extension, and medial rotation of the shoulder.
- ❑ Lesion of Thoracodorsal nerve significantly affects extension

Sensory deficit:

- No sensory loss



 **Compete the following table with appropriate nerves suitable for the abnormalities that occur when these nerves are injured ?**

Deformity	Nerve
Flatting of shoulder contour	Axillary
Wrist drop	Radial
Partial Claw hand	Ulnar
Benediction hand	Median
Winging of scapula	Long thoracic
Ape like hand	Median
Waiter's tip position	Upper trunk of brachial plexus Erb's Douchenne paralysis
Combined Claw hand	Lower trunk of brachial plexus Klumpke's palsy

Humeral fractures

Surgical Neck Fracture of Humerus

Case scenario: A 70-year-old woman presents after a **fall onto her outstretched hand** with pain and inability to lift her arm. Exam reveals ecchymosis over the proximal arm, crepitus on motion, and **intact radial pulse**

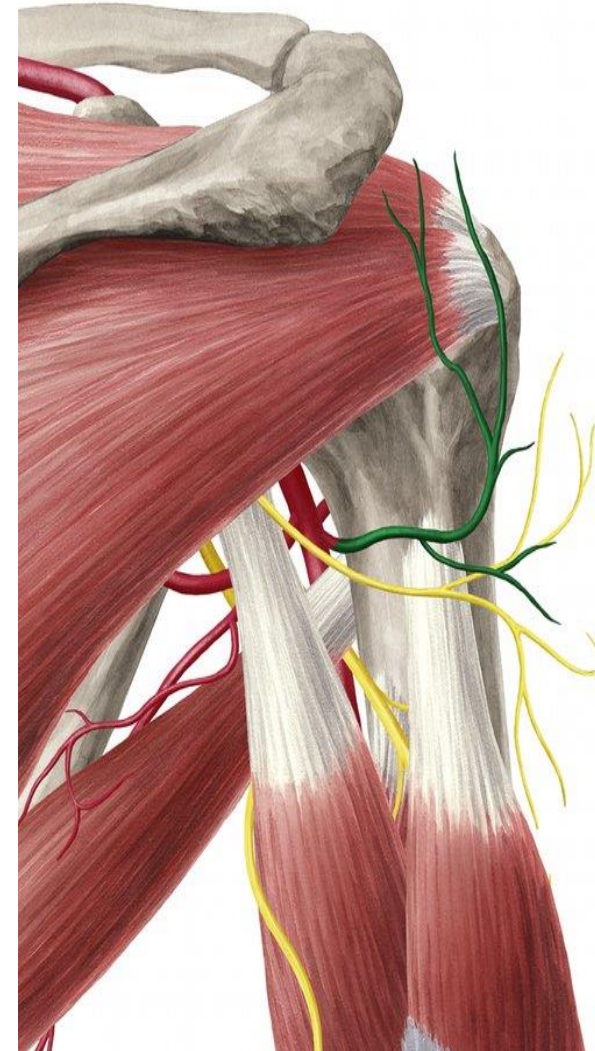


Surgical Neck Fracture of Humerus

Key Features:

- **High-risk population:** Elderly (osteoporosis), seizures, electrocution.
- **Critical structures at risk:**
 - Axillary nerve (runs 5 cm distal to acromion).
 - Posterior circumflex humeral artery (adjacent to nerve).
- **Deformity:** Proximal fragment abducted/externally rotated (pectoralis major pull)

Why "surgical": High risk for neurovascular injury (vs. "anatomical" neck).



Midshaft Humeral Fracture

Case scenario: 35-year-old construction worker falls from scaffolding, landing on his right arm. He presents with **visible deformity at mid-arm**, **weak wrist extension**, and **numbness over the dorsum of the hand**.



Surgical Neck Fracture of Humerus

Key Features:

- **"Hanging arm" deformity**: Dependent distal fragment due to gravity.
- **Radial nerve jeopardy**: Wraps around spiral groove (10-14 cm distal to acromion)

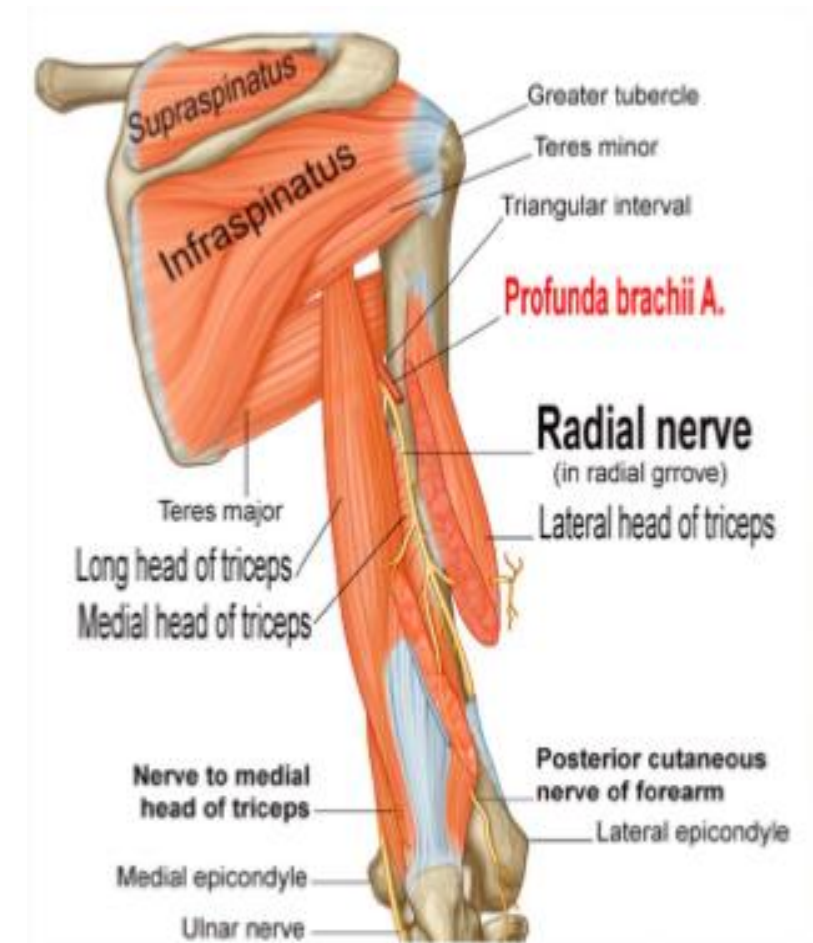
Anatomical features

1. Radial Nerve Course:

- **Proximal to fracture**: Innervates triceps (often spared in midshaft fractures).
- **At spiral groove**: Vulnerable to laceration/stretch (test: wrist/finger extension).
- **Distal to fracture**: Emerges lateral to brachioradialis (sensory: dorsal hand).

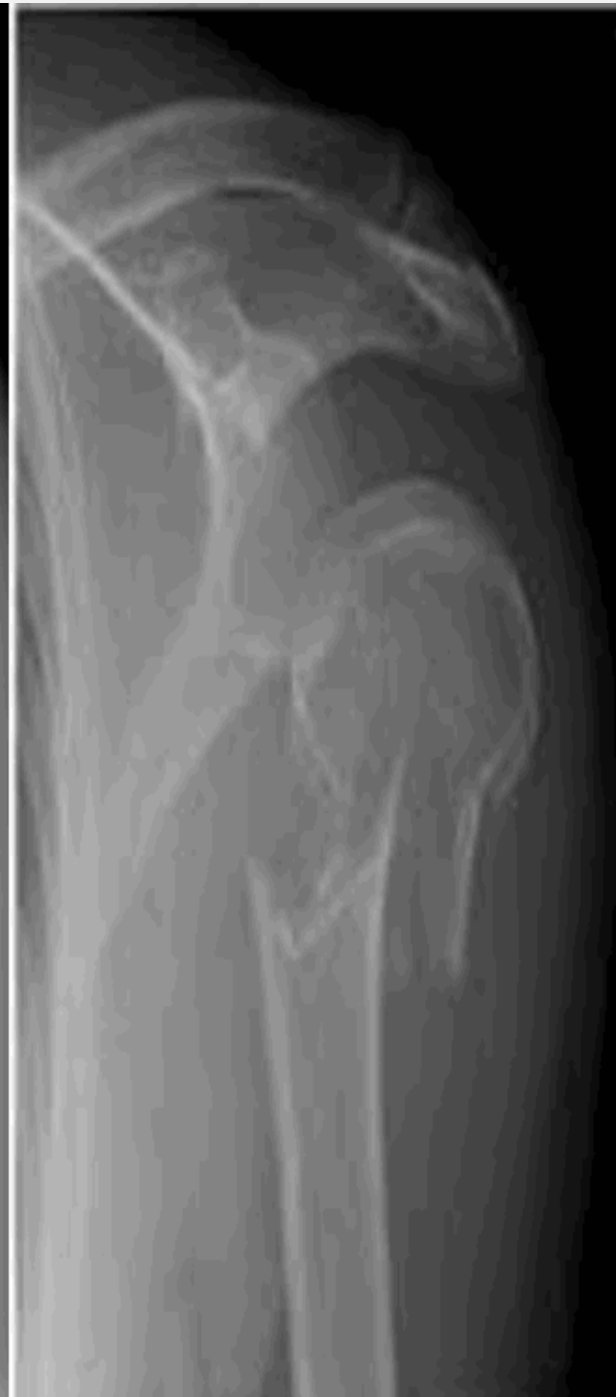
2. Muscle Forces & Displacement

- Proximal fragment: Pulled anteriorly by pectoralis major.
- Distal fragment: Pulled medially by brachialis/biceps.



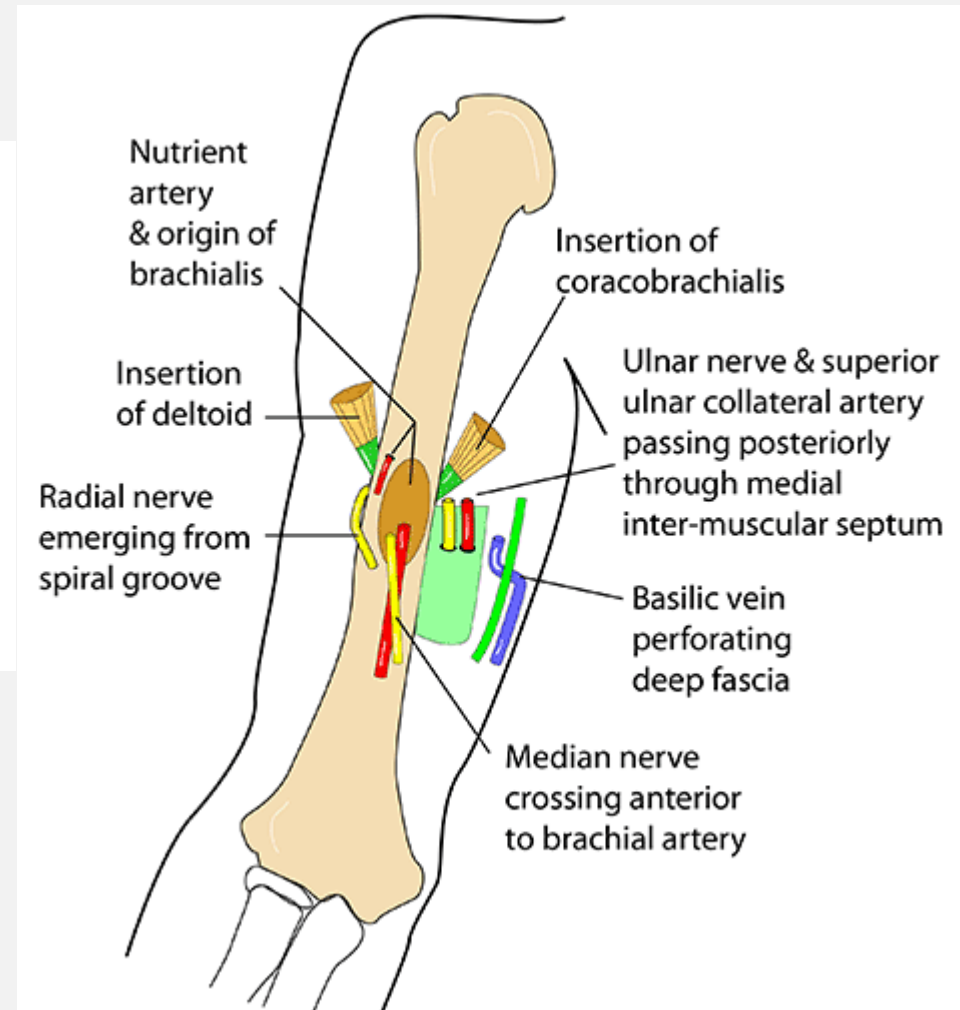
□ in fractures of humeral shaft, the fractured segments displaced depends on sit of fracture. Explain how is that?

Site of fracture	Proximal segment	Distal segment
Above bicipital groove	Abducted by muscles attached to greater tuberosity(mainly supraspinatus)	Adducted by muscles attached to bicipital groove
Above the level of deltoid tuberosity	Adducted by muscles attached to bicipital groove	Abducted & pulled up by deltoid muscle
Below the level of deltoid tuberosity	Abducted by deltoid	Adducted & pulled up by biceps & triceps



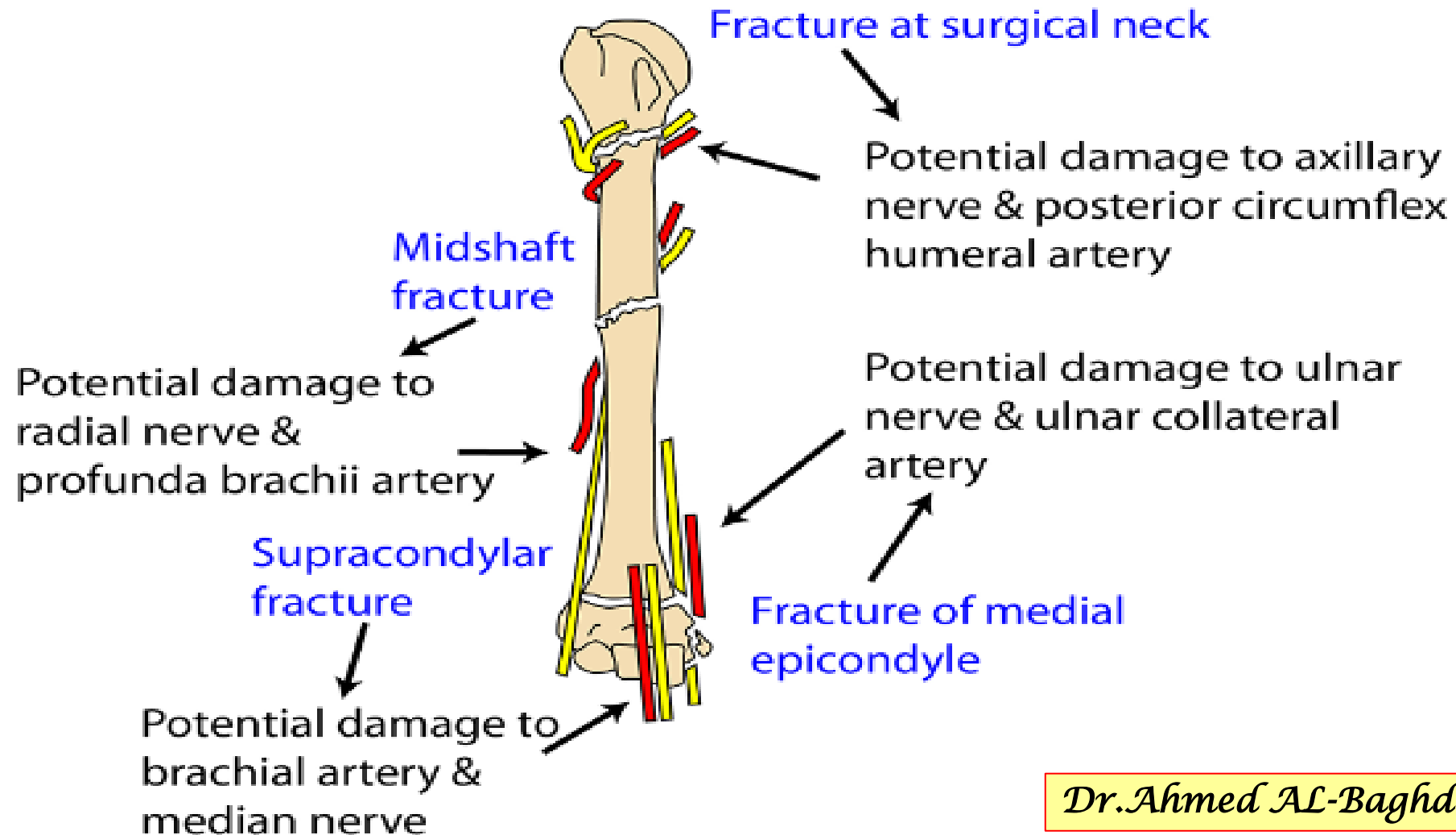
□ what are the anatomical events that occur at or near mid-shaft of humerus?

- Nutrient artery entering humerus
- Origin of brachialis
- Insertion of deltoid
- Insertion of coracobrachialis
- Radial nerve emerging from spiral groove
- Basilic vein perforating deep fascia
- Median nerve crossing anterior to brachial artery
- Ulnar nerve & superior ulnar collateral artery passing posteriorly through medial inter-muscular septum



□ What are the structures in direct contact with humerus and liable for injury?

ARTERIES AND NERVES LIABLE TO DAMAGE AT 4 SITES OF FRACTURE OF THE HUMERUS



Pediatric Supracondylar Humerus Fractures

Case scenario: 6-year-old boy falls from monkey bars onto an **outstretched hand**, presenting with a swollen, **painful elbow held in extension**. Exam reveals no **radial pulse but warm, pink fingers**



Pediatric Supracondylar Humerus Fractures

Key Features:

- **High-risk population:** Ages 5-7 years (weak supracondylar area).
- **Critical structures at risk:**
 - Brachial artery (pulse loss in 5-15% of type III fractures).
 - Median/radial/ulnar nerves (anterior interosseous nerve most commonly injured).
- **"Snapping pop" sign:** Audible crack during injury suggests severe displacement

Pediatric Supracondylar Humerus Fractures

Anatomical features

Supracondylar Region:

- o Thin cortex: Weakest point between coronoid/olecranon fossae.
- o Growth plates: Distal humerus physis contributes 80% of elbow growth.

Neurovascular Relations:

- o Brachial artery: Anteromedial to fracture; pulse loss $\neq$ ischemia (collaterals may preserve hand perfusion).

Nerves:

- o Anterior interosseous (AIN): Test with "OK sign" weakness.
- o Radial: Test thumb/wrist extension.
- o Ulnar: Test finger abduction/intrinsic function.

Displacement Forces:

- o Extension-type (95%): Distal fragment displaces posteriorly (triceps pull).
- o Flexion-type (5%): Distal fragment displaces anteriorly (rare, higher neurovascular risk).

Pediatric Supracondylar Humerus Fractures

Anatomical features

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Compartment syndrome risk:

Volkmann's ischemic contracture if missed vascular injury.

❑ What is Volkmann's ischemic contracture?

In supracondylar fracture

- There is displacement of proximal bone fragment forward & downward, compressing brachial a. & median n.
- When the collateral circulation cannot compensate for occlusion of the brachial artery leads to ischemia of flexor muscles of forearm & necrosis
- the muscles replaced by fibrous tissue leading to thinning & shortening of the muscle
- result in flexion deformity of wrist & fingers **(Volkmann's contracture)**





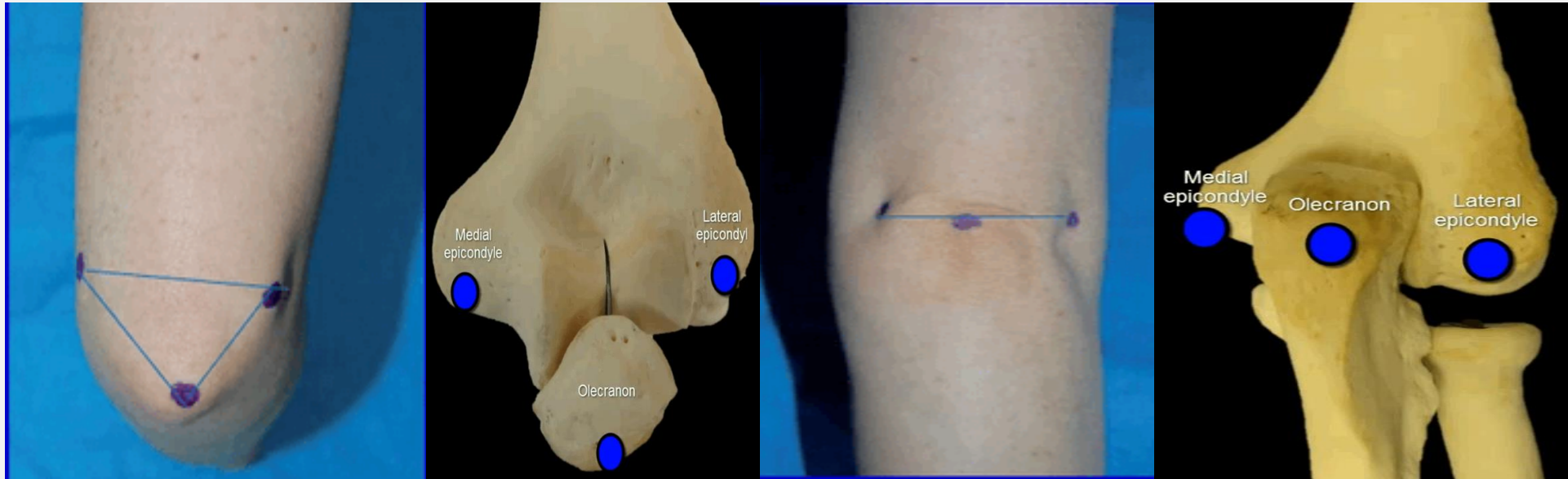
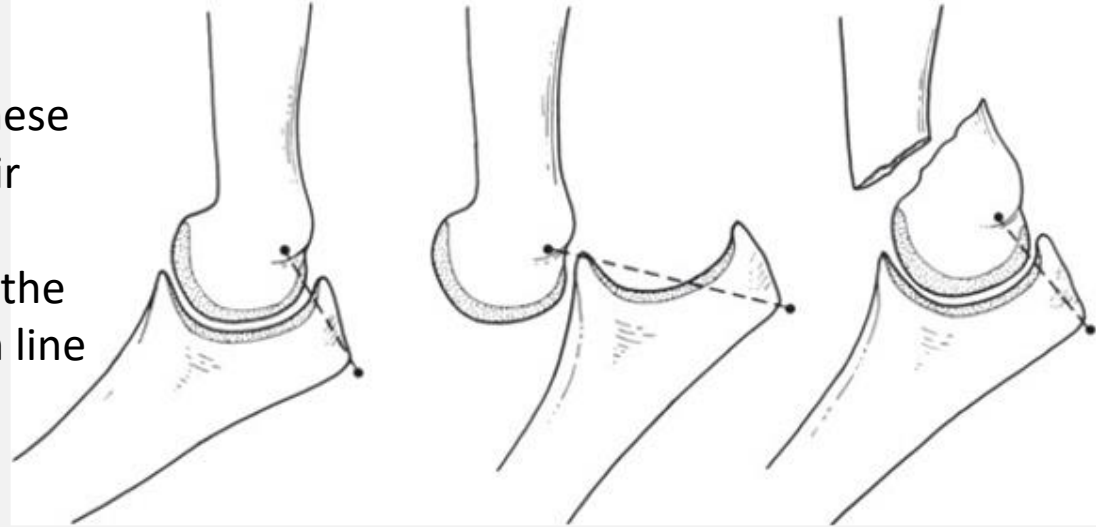
Fractures of the humerus

nerve	Site of contact	deformity
Axillary	Around surgical neck	Flat shoulder
Radial	Spiral groove	Wrist drop
ulnar	Behind medial epicondyle	Claw hand



□ why in supracondylar fracture of humerus ,triangle relationship of the 3 bony points(olecranon, medial & lateral epicondyles) is not disturbed but in elbow dislocation , it is disturbed?

- A supracondylar fracture lies above these points, which therefore remain in their triangular relationship to each other
- in dislocation of the elbow, however, the olecranon comes to lie more or less in line with the epicondyles



Radial Head Subluxation (Nursemaid's Elbow)

Case scenario: A 2-year-old girl presents with **sudden refusal to use her left arm** after her mother **pulled her up by the wrist**. The arm is held in **slight flexion and pronation** with **no swelling or deformity**



Radial Head Subluxation (Nursemaid's Elbow)

Key Features:

- **Classic mechanism:** Axial traction on a pronated forearm (e.g., lifting child by hands).
- **Pathognomonic finding:** No voluntary supination (child won't reach for toys with affected hand).
- **Age window:** Peak incidence 1-4 years (ligamentous laxity + immature radial head).

○ Anatomical features

Radial Head Development:

- <5 years old: Radial head is immature narrow ovoid (not fully spherical) → easier subluxation.
- Annular ligament: Thin, elastic and loose in toddlers; slips into radiocapitellar joint with traction.

Subluxation Mechanism:

- Pronation + Traction → annular ligament slips over radial head → trapped in joint.
- Reduction: Supination + flexion (lifts ligament out of joint).

It's Unique to Kids:

Adult equivalent: Elbow dislocation (ligaments tear vs. stretch in children).

- Differential: Pulled elbow vs. fracture (no tenderness over radial head in subluxation)

